Modelling Conditional First and Second Moment of Univariate Cryptocurrency Return Distributions: A GARCH Survey

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GitHub: `https://github.com/MDrschl/VolatilityModelling`

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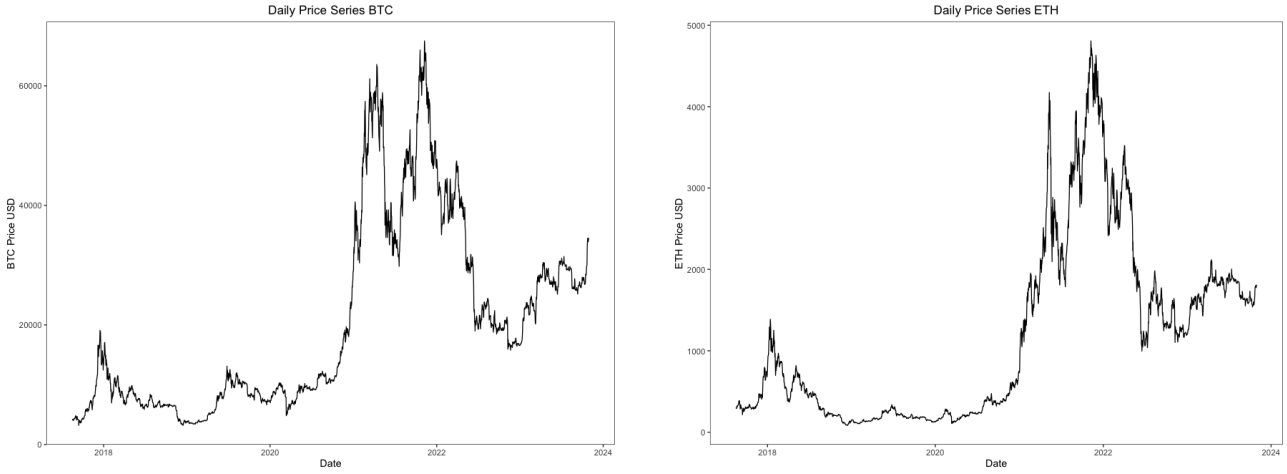
1 Introduction

During the past decade, cryptocurrencies have evolved from a fringe digital curiosity to a globally traded liquid asset with growing derivative markets. As market participants seek to hedge, speculate, or arbitrate amid their pronounced volatility, crypto options and futures have become valuable tools for expressing both directional views and uncertainty. In this assignment, we build on the findings of Chi and Hao (2021) that a parsimonious single-variate GARCH(1,1) model outperforms both richer multivariate DCC-GARCH specifications and option-implied volatilities in forecasting future realized volatility. We extend their framework in several ways and lengthen the estimation window. Whereas Chi & Hao focus on data roughly spanning 2016-2020, we use Bitcoin and Ethereum returns from January 2017 through October 2023 as our in-sample period, then reserve November 2023 through April 2025 for out-of-sample comparison. This much longer estimation horizon allows us to assess model stability across distinct market regimes while keeping aside the most recent crypto-cycle to evaluate true forecast performance.

We investigate the characteristics of the first and second conditional moments of univariate return distributions for major cryptocurrencies by estimating a broad class of ARMA-GARCH-type models across multiple sampling frequencies. To account for dynamic co-movements and conditional correlations, we further employ a DCC-GARCH framework to model multivariate effects between assets.

2 Data

To conduct our analysis, we collected high-frequency price data for Bitcoin (BTC) and Ethereum (ETH) directly from the Binance exchange using their public REST API. Specifically, we use 1-minute OHLCV (Open, High, Low, Close, Volume) data for the BTCUSDT and ETHUSDT pairs. The dataset spans from August 17, 2017, to May 1, 2025 and covers over seven years of uninterrupted trading history. While our primary focus lies on modelling the volatility of each asset individually, we also investigate the degree of co-movement of BTC and ETH across different frequencies. Figure 1 displays the daily price evolution of BTC and ETH over the sample period. The visual similarity in their major peaks and drawdowns supports a hypothesis of structural co-movement between the two assets. We expect constant or dynamic correlations between the return series. Capturing and modelling dynamic conditional correlations between the standardised residuals of the two return series enhances the quality of volatility forecasts by allowing past shocks in Bitcoin to contain predictive information about future shocks in Ethereum, and vice versa.



(a) BTC Daily Price Series

(b) ETH Daily Price Series

Figure 1: Daily closing price series for BTC and ETH from 2017 to 2024

3 Methodology

Our methodology begins with a descriptive analysis of high frequency Bitcoin and Ethereum return data. We compute robust autocorrelation and partial autocorrelation functions using confidence bands adjusted for heteroskedasticity. Finally, we derive realized volatility by aggregating one minute returns into hourly, six hour and daily intervals, producing frequency specific variance measures that allow us to compare volatility dynamics across multiple time scales.

To model and forecast volatility, the analysis primarily focuses on univariate approaches, assuming diagonal matrices $\mathbf{A}_j$ and $\mathbf{B}_j$ (cf. Chapter 5, Lecture Notes). This specification enables detailed model construction and facilitates parameter interpretation at the individual asset level. Accordingly, several GARCH-type models are estimated, including standard GARCH (sGARCH; (Bollerslev, 1986)), threshold GARCH (TGARCH), exponential GARCH (eGARCH; (Nelson, 1991)), Glosten-Jagannathan-Runkle GARCH (GJR-GARCH; (Glosten et al., 1993)), and Markov-switching GARCH (MS-GARCH; (Gray, 1996)). To build on the findings of Chi and Hao (2021), the model set is further extended to incorporate regime-switching dynamics. As highlighted in their discussion of volatility modelling, regime-switching structures are particularly relevant for cryptocurrency return series, where distinct high- and low-volatility regimes are frequently observed, warranting their inclusion in future empirical research.

To further extend our analysis to the multivariate setting, we are employing the DCC-GARCH model (Engle, 2002) to capture potential time-varying co-movements in the shocks of the two return series under consideration. The models explicitly account for cross-asset dependence and spillover effects, which are especially relevant in joint portfolio or risk management contexts. Note, however, that this framework does not allow for the explicit modelling of interaction effects, such as how shocks to BTC influence the volatility of ETH and vice versa. Chi and Hao (2021) addressed this limitation by employing a DCC-MGARCH model, where the conditional correlation matrix of the shocks is modelled dynamically while the multivariate GARCH component captures cross-dependencies in the conditional variance structure. Despite this, the DCC-GARCH model still enables us a meaningful analysis of the conditional covariance structure between BTC and ETH by capturing the evolution of their time-varying correlations.

We then perform a comprehensive grid search over combinations of ARMA(p,q)-GARCH(r,s) specifications, where in each case we assume a Student-t distribution for the shocks. To select the best model, we minimize the Bayesian Information Criterion (BIC). We choose the BIC because it balances model fit against complexity by penalizing additional parameters. This is particularly important when modelling Bitcoin and Ethereum volatility to avoid overfitting short-lived fluctuations and to enhance out-of-sample forecast performance which contrasts to the Akaike Information Criterion (AIC) which favours more complex specifications which improve in sample fit but risk overfitting. Since conditional mean and variance cannot be estimated jointly in the Markov switching GARCH, for simplicity we fit an AR(1) process (which always yielded the lowest joint BIC for the remaining GARCH specifications) and then apply the MS-GARCH filter to its residuals. Once the optimal model order is identified, we apply the Ljung Box test to its standardized residuals to verify the absence of remaining linear autocorrelation, the McLeod Li test to confirm persisting ARCH effects in the squared standardized residuals. To further inspect the distribution of the standardized residuals from the ARMA-GARCH specifications, we plotted Q-Q plots.

Once the in-sample model selection based on daily, 6-hourly, and hourly returns between August 17, 2017, and October 31, 2023, is completed, the analysis proceeds with the out-of-sample forecasting comparison, which is based on daily returns from November 1, 2023, to April 30,

2025. The focus on daily returns in the out-of-sample period is motivated by two considerations. First, generating volatility forecasts on a rolling basis using hourly data proved computationally infeasible, as the models require refitting on relatively large samples to maintain predictive accuracy. Second, as discussed with the supervisor, the original objective of the volatility forecasts was to compute multi-day average volatility estimates up to the expiration of European call and put options on the respective crypto assets. These estimates would then be compared to the implied volatility of the corresponding options to construct a volatility spread strategy. For this purpose, daily return-based forecasts offer sufficient granularity to support the strategy's design.

We assess predictive performance using rolling volatility forecasting. Rolling volatility forecasts are generated using the best-performing models identified in-sample via model selection criteria. These forecasts are then compared to realized volatility estimates to evaluate out-of-sample accuracy and assess each model's ability to capture the asymptotically true variability of returns under different market conditions. Specifically, we use scatter plots, time series plots, Mincer-Zarnowitz regressions, the SPA test, and the Model Confidence Set (MCS) procedure.

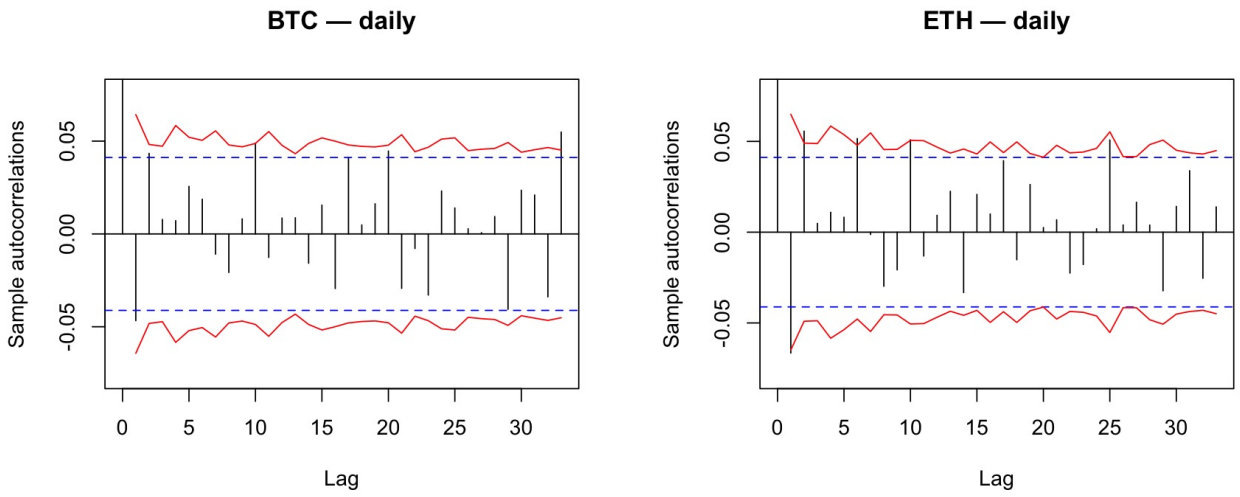
4 Descriptive Statistics

Table 1 presents key descriptive statistics for BTC and ETH returns across three sampling frequencies. On average, ETH exhibits slightly higher returns than BTC across all frequencies, particularly at the daily level. As the sampling frequency increases, volatility increases, with hourly returns showing the highest annualized volatility, most notably ETH, at over 81%. Skewness is mildly negative across most series, indicating slight asymmetry toward negative returns, though some mid-range frequencies show positive skewness. Excess kurtosis rises sharply at higher frequencies, especially for BTC and ETH hourly returns, reflecting heavy tails and an increased likelihood of extreme observations. These distributional features contrast with those typically found in equity markets, such as the S&P500, where volatility tends to respond more strongly to negative returns due to the leverage effect. In cryptocurrencies, this asymmetry is largely absent or reversed, suggesting a different risk structure and investor response dynamic.

Table 1: Descriptive Statistics by Time Horizon and Asset

Asset	**Time Horizon**	**Average Return (%)**	**Annualized Volatility (%)**	**Skewness**	**Excess Kurtosis**
BTC	1 day	0.169	61.78	-0.282	4.820
BTC	6 hours	0.042	43.82	-0.084	7.586
BTC	1 hour	0.008	66.99	-0.027	28.114
ETH	1 day	0.202	77.99	-0.271	3.109
ETH	6 hours	0.049	53.85	-0.202	4.769
ETH	1 hour	0.009	81.67	-0.146	21.123

Figure 2 presents the robust autocorrelation function (ACF) plots for the daily return series of BTC and ETH. The plots display the sample autocorrelations along with standard and robust confidence bands. We observe persistent and statistically significant autocorrelations at several lag orders that exceed the 95% confidence bounds. This evidence of linear dependence in the level series motivates our decision to first model the conditional mean with an $ARMA(p, q)$ specification, thereby capturing predictable return dynamics before estimating volatility. By fitting an ARMA process to the returns, we prevent GARCH from conflating mean-level autocorrelation with volatility clustering, allowing its parameters to be estimated on genuinely unpredictable shocks. We apply the same procedure at six-hourly and hourly frequencies and obtain similar results, confirming that our ARMA–GARCH approach is valid across all sampling horizons. Consequently, the fitted σ_t^2 will capture how volatility shocks persist over time, reflecting true risk dynamics, such as volatility clustering and leverage effects, rather than residual autocorrelation in the return mean.



(a) BTC Daily Robust ACF

(b) ETH Daily Robust ACF

Figure 2: Robust ACF plots for BTC and ETH daily return series

Table 2 reports the results of three diagnostic tests on BTC and ETH returns at daily, six-hour, and hourly frequencies. The Ljung–Box statistics are very large across all horizons, for example 43,475.85 for BTC daily ($p = 0$) and 43,459.69 for ETH daily ($p = 0$), indicating strong serial correlation in the return levels. The McLeod–Li statistics rise from 41,879.30 for BTC daily ($p = 0$) and 41,264.88 for ETH daily ($p = 0$) to over one million for both assets at the hourly frequency, confirming the presence of time-varying volatility. The Jarque–Bera statistics also increase sharply as sampling becomes finer, moving from 363.55 for BTC daily ($p = 0$) and 452.93 for ETH daily ($p = 0$) to 8,722.85 and 10,807.36 for BTC and ETH hourly respectively (both $p = 0$), demonstrating persistent departures from normality with pronounced skewness and kurtosis. Together, these results firmly reject the null hypotheses of independence, homoskedasticity, and normality. These findings align with the stylised fact of volatility clustering, suggesting that GARCH models could fit well.

Table 2: Diagnostic Test Results for BTC and ETH Return Series

Series	Asset	Ljung-Box Stat	Ljung-Box p	McLeod-Li Stat	McLeod-Li p	Jarque-Bera Stat	Jarque-Bera p
BTC Daily	BTC	43,475.85	0	41,879.30	0	363.55	0
BTC 6-Hour	BTC	179,367.83	0	177,670.81	0	1,456.53	0
BTC Hourly	BTC	1,083,177.00	0	1,081,311.98	0	8,722.85	0
ETH Daily	ETH	43,459.69	0	41,264.88	0	452.93	0
ETH 6-Hour	ETH	179,280.29	0	176,801.36	0	1,806.18	0
ETH Hourly	ETH	1,083,118.23	0	1,080,363.84	0	10,807.36	0

5 In-Sample Evaluation

In this study, we conduct a comprehensive comparison of several univariate GARCH-type models to evaluate their in-sample performance on BTC and ETH return series across two different frequencies: daily and 6-hourly. We recognize that financial return data, in particular in cryptocurrency markets, often exhibit low but non-zero mean returns and short-term autocorrelation. As outlined in the previous chapter, the asset returns on the squares exhibit strong autocorrelation patterns. Additionally, they exhibit conditional heteroscedasticity. This implies that the volatility of returns varies over time, and there is a tendency for clustering, where large returns are followed by similarly large returns of either sign. A broad stream of literature proposes AR-GARCH models to properly deal with these features.

We use the standard ARMA-GARCH model as our baseline, which captures symmetric volatility clustering through past squared residuals and lagged conditional variances. In order to address certain asymmetric volatility responses, we consider three extensions: the Threshold GARCH (TGARCH), which incorporates a dummy-based asymmetry term; the GJR-GARCH, which captures leverage effects via an indicator function on negative shocks; and the Exponential GARCH (EGARCH), which models log-volatility and allows for both magnitude- and sign-dependent asymmetries without imposing positivity constraints.

In addition to these classical models, we explore two advanced specifications: a univariate Markov-Switching GARCH (MS-GARCH), and a multivariate Dynamic Conditional Correlation GARCH (DCC-GARCH), whose univariate margins provide valuable insights into time-varying volatility behaviour while the conditional correlations will help us in assessing the co-movements of the two asset return distributions. Throughout our analysis, we focus on how the structural differences and parameterizations of each model impact their ability to capture the stylized facts of crypto-asset returns, including volatility clustering, asymmetry, and regime dependence.

5.a. Univariate

We first examine five univariate volatility models: GARCH, GJR-GARCH, T-GARCH, eGARCH, and MS-GARCH. For each model type, a grid search over ARMA-GARCH specifications was conducted to identify the optimal model order based on the Bayesian Information Criterion. For the daily return series of both BTC and ETH, the best-performing specification across all model classes was consistently the AR(1)-GARCH(1,1) model with Student-t distributed innovations. This result aligns with the findings of Chi and Hao (2021), who also identify this specification as optimal for daily cryptocurrency returns. It further confirms the broader literature on equity return series, where the AR(1)-GARCH(1,1) with Student-t innovations is a widely used benchmark.

In contrast, the results for six-hour return series differ across assets and model types. For BTC, the AR(1)-GARCH(2,1), AR(1)-GJR-GARCH(1,2), and AR(1)-eGARCH(3,3) specifications yield the best in-sample performance depending on the model class. For ETH, the AR(1)-GJR-GARCH(2,2) and AR(1)-eGARCH(3,3) specifications are found to be superior. These findings suggest that higher-frequency data may require more flexible or higher-order volatility structures to adequately capture the dynamics in return volatility, and that the optimal specification may be asset-specific at intraday frequencies.

In the following sections, the discussion will focus on the process specifications and the identification of the best-performing in-sample models, which will then serve as the basis for the subsequent out-of-sample comparison.

For the daily standard GARCH models, we observe both a statistically significant α and β

values for both Bitcoin and Ethereum which confirms that volatility is driven by both recent return shocks and its own past levels, resulting in pronounced clustering and persistence in risk. However, at the 6 hour level, while the β values remain significant for the Bitcoin series, this is not the case for Ethereum. Furthermore, we observe a divergence in volatility dynamics between Bitcoin and Ethereum. Bitcoin's best-fitting specification is a GARCH(1,2), in which both lagged-variance coefficients β_1 and β_2 are statistically significant. This indicates that multiple past variance terms contribute to today's volatility, suggesting a more complex persistence structure regarding intraday horizons. By contrast, Ethereum's six-hour series is best described by a GARCH(1,1) in which neither the ARCH coefficient α nor the GARCH coefficient β is significant. In other words, six-hour Ethereum volatility behaves nearly like unpredictable noise once standard GARCH effects are controlled for. These findings imply that Bitcoin's intraday risk retains longer-memory effects that warrant richer GARCH specifications, whereas Ethereum's six-hour variance may be dominated by microstructure noise or exogenous drivers not captured by simple GARCH terms.

The asymmetric GARCH models we estimated all attempt to capture a "leverage effect," whereby negative shocks raise future volatility more than positive shocks. In our daily TGARCH fits, the threshold coefficient η is both small (around -0.02 for Bitcoin and -0.02 for Ethereum) and statistically insignificant (p-values above 0.4). The GJR-GARCH leverage term γ likewise fails to attain significance in either series. In contrast, for the EGARCH(1,1) specifications, the estimated asymmetry coefficients γ_1 are positive and statistically significant for both Bitcoin ($\hat{\gamma}_1 = 0.1969$, $p < 0.001$) and Ethereum ($\hat{\gamma}_1 = 0.1943$, $p = 0.0107$), indicating that positive shocks tend to raise conditional variance more than negative shocks. In the first section of Table 3 the parameter estimates for the AR(1)-EGARCH(1,1) with Student-t distributed innovations are provided. This contrasts with the negligible leverage effects in TGARCH and GJR-GARCH and suggests that, while cryptocurrencies may lack the traditional balance-sheet leverage mechanism, they still exhibit an asymmetric volatility response. Together these results imply that cryptocurrency volatility does not respond asymmetrically to past returns in the way equity volatility famously does. The absence of a meaningful leverage effect suggests that the balance-sheet channel is not operative in markets for digital assets. From a modeling perspective this validates the use of symmetric GARCH(1,1) specifications. They achieve equivalent in-sample fit without the added complexity of asymmetric terms, and they avoid overfitting noise in the absence of a true downside-shock channel. Like Chi and Hao , we find negligible leverage effects in TGARCH and GJR-GARCH and only marginal asymmetry in EGARCH.

The two-regime eGARCH(1,1) fits for daily Bitcoin and Ethereum returns reveal similar state-dependent dynamics with notable differences in tail behavior and persistence. For Bitcoin, Regime 1 is characterized by modest ARCH and very high GARCH coefficients ($\alpha_1 = 0.0529$, $\beta_1 = 0.9952$) and nearly Gaussian tails ($\nu_1 \approx 99.5$), whereas Regime 2 exhibits a substantially larger ARCH term ($\alpha_2 = 0.2055$), slightly lower but still high GARCH persistence ($\beta_2 = 0.9641$) and fat tails ($\nu_2 \approx 5.04$). By contrast, Ethereum's regimes (Table A21) both display heavy-tailed behavior but differ in persistence: Regime 1 has $\alpha_1 = 0.2072$, $\beta_1 = 0.9934$ and $\nu_1 \approx 3.06$, while Regime 2 shows a smaller ARCH coefficient ($\alpha_2 = 0.3379$), much lower GARCH persistence ($\beta_2 = 0.7211$) and thinner tails ($\nu_2 \approx 35.4$). The transition probabilities ($p_{11} \approx 0.44$, $p_{22} \approx 0.37$ for Bitcoin; $p_{11} \approx 0.83$, $p_{22} \approx 0.54$ for Ethereum) imply that Bitcoin switches frequently between calm and turbulent states, whereas Ethereum remains in its calm regime more persistently. We can therefore infer that Bitcoin alternates unpredictably between tranquil and highly volatile periods, while Ethereum tends to linger longer in its low-volatility state before transitioning to turbulence. This may reflect that Bitcoin reacts more sharply to changes in market sentiment, while Ethereum's volatility dynamics are comparatively smoother.

5.b. Multivariate

As the AR(1)-EGARCH(1,1) specification has been identified as a suitable in-sample model for the daily return series of both BTC and ETH, it will be employed as the univariate conditional variance model for each asset within the DCC framework used to estimate the time-varying covariance matrix. As in the univariate case, the optimal DCC order and the most appropriate assumption regarding the multivariate distribution were identified by iterating over different model specifications and selecting the one with the lowest BIC. The best model is the DCC(1,1) with a multivariate Student-t distribtion.

The estimated DCC parameters provide insight into the dynamic correlation process between BTC and ETH returns. The parameter $dcca_1 = 0.0651$ captures the short-term influence of past standardized shocks on the conditional correlation. Its statistical significance (t-value $= 5.879$, $p < 0.0001$) indicates that recent innovations in returns have a meaningful impact on the evolution of correlations. The parameter $dccb_1 = 0.9289$ reflects the persistence of the correlation dynamics. Its magnitude, being close to one and highly significant (t-value $= 69.473$, $p < 0.0001$), suggests that the conditional correlations are highly persistent and adjust slowly over time. Furthermore, the estimated joint degrees of freedom parameter $mshape = 4.0000$, with a t-value of 8.837, confirms the relevance of using a multivariate Student-t distribution to model the heavy tails observed in the joint return distribution. Overall, the estimates support a flexible correlation structure that is both reactive to shocks and capable of capturing non-Gaussian features in the joint behavior of BTC and ETH returns. These findings become even more evident when examining Figure 3, which displays the off-diagonal elements of the conditional correlation matrix. The correlations are clearly strongly positive and relatively persistent. However, an interesting observation is that the conditional correlations appear to have become more stable in recent years, as fluctuations have decreased both in frequency and magnitude. In the study by Chi and Hao (2021), the estimated value of $dcca_1$ was slightly higher and the $dccb_1$ parameter slightly lower. This is consistent with their sample period, which covers 2017 to 2020, a time span during which greater volatility in conditional correlations is apparent.

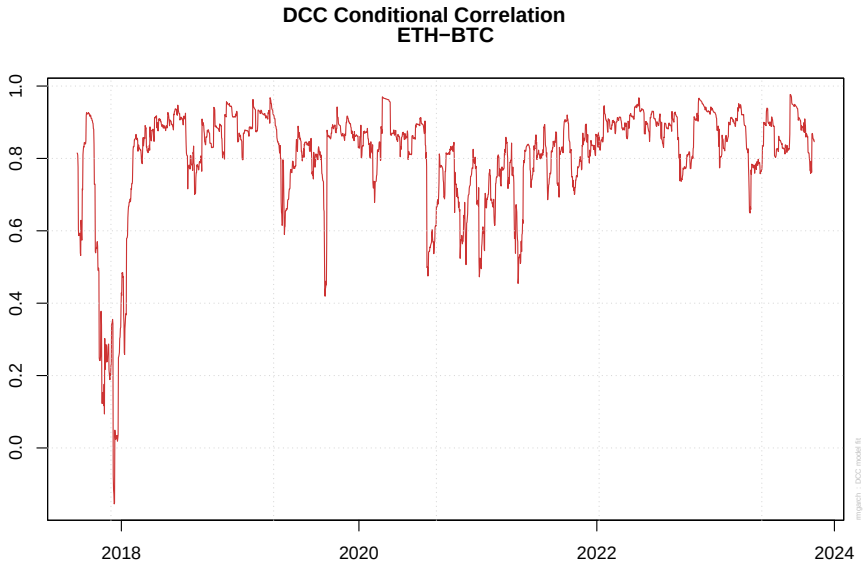


Figure 3: Conditional Correlation of Daily Returns from a DCC(1,1)-AR(1)-GARCH(1,1) with Student-t innovations

Parameter	Estimate	Std. Error	t value	Pr(>|t|)
BTC Series				
[BTC] μ	0.000723	0.000407	1.775	0.0758
[BTC] AR(1)	-0.064438	0.016451	-3.917	< 0.0001
[BTC] ω	-0.018521	0.006436	-2.878	0.0040
[BTC] α_1	0.005467	0.013706	0.399	0.6900
[BTC] β_1	0.996841	0.001188	839.196	< 0.0001
[BTC] γ_1	0.196920	0.017333	11.361	< 0.0001
[BTC] ν	2.664592	0.256308	10.396	< 0.0001
ETH Series				
[ETH] μ	0.000754	0.000627	1.203	0.2288
[ETH] AR(1)	-0.096577	0.019175	-5.037	< 0.0001
[ETH] ω	-0.077423	0.043003	-1.800	0.0718
[ETH] α_1	0.002875	0.017745	0.162	0.8713
[ETH] β_1	0.986870	0.007638	129.204	< 0.0001
[ETH] γ_1	0.194303	0.081024	2.398	0.0165
[ETH] ν	3.174228	0.388080	8.179	< 0.0001
DCC Parameters				
$dcca_1$	0.065077	0.011070	5.879	< 0.0001
$dccb_1$	0.928877	0.013370	69.473	< 0.0001
mshape (Joint ν)	4.000000	0.452661	8.837	< 0.0001
Model Fit				
Log-Likelihood (ℓ)		9891.15		
Number of Parameters (k)		18		
AIC		-19765.30		
BIC		-19650.96		
Diagnostic p-values				
Ljung-Box (z_t^2)		0.505		
Ljung-Box (z_t)		0.0017		

Table 3: Daily AR(1)-eGARCH(1,1) DCC-GARCH Estimation with Student-t Innovations

5.c. Final model selection

After discussing the process characteristics and identifying the best-performing model orders based on the lowest BIC values, we proceed by selecting models for the out-of-sample comparison measured on daily return distributions. This choice is motivated by previously stated concerns regarding the appropriateness of conducting out-of-sample comparisons at higher frequencies. Although the AR(1)-sGARCH(1,1) model does not exhibit the best in-sample performance relative to more flexible specifications, it is included in the following analysis for two reasons: first, it serves as a widely used benchmark model; and second, it demonstrated relatively strong out-of-sample performance in the study by Chi and Hao (2021) for both assets. Similarly, the AR(1)-eGARCH(1,1) model consistently outperforms alternative univariate models for both BTC and ETH. In addition, the regime-switching model is included despite not being the best in-sample model according to BIC values. Its inclusion is justified by its potential to provide valuable insights in the out-of-sample setting and to serve as an empirical extension to the findings of Chi and Hao (2021).

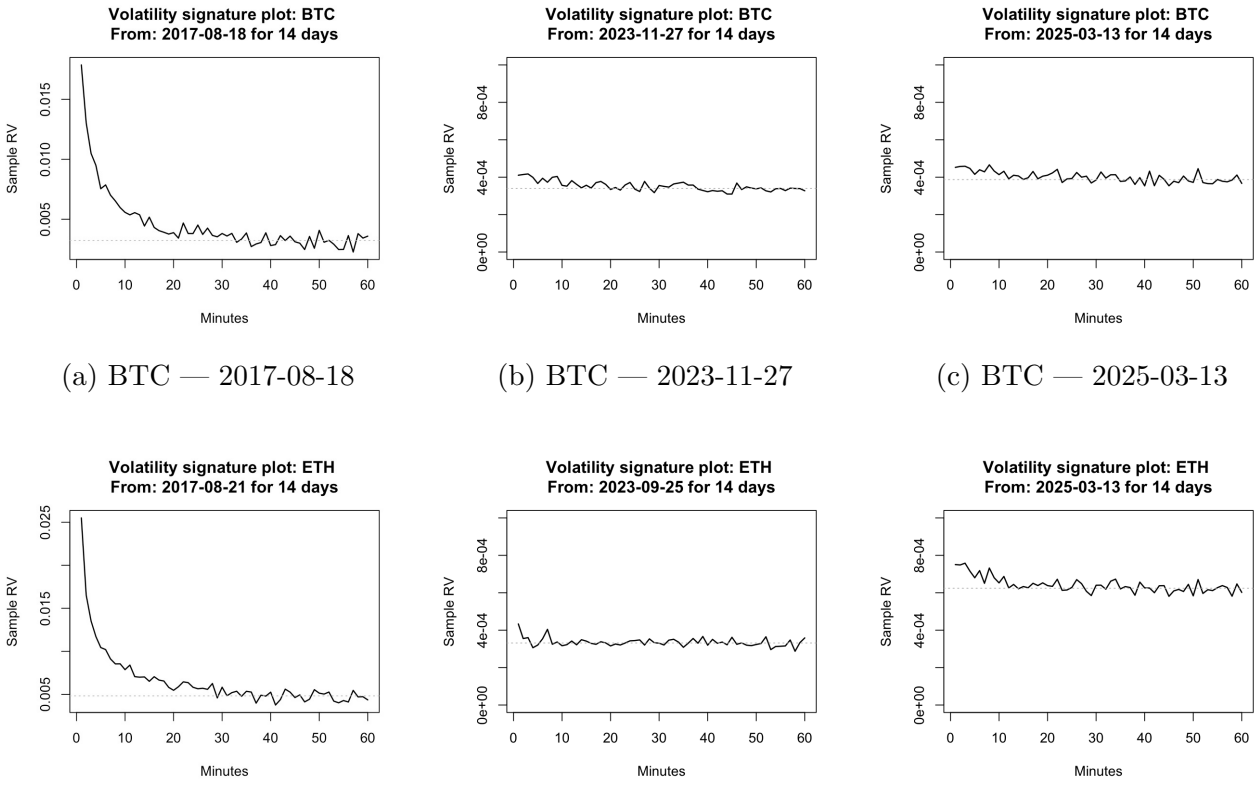
6 Out-of-Sample Evaluation

6.a. Volatility Forecasts

In a first step to compare the out-of-sample performance of the different models selected in the previous section was to compute the conditional volatility forecasts. We implemented The rolling forecast procedure estimates conditional volatility using an ARMA-GARCH model with a one-step-ahead prediction horizon. For each time point $t+1$ in the out-of-sample period, the model is re-estimated using all return observations up to time t. A one-step-ahead forecast of the conditional standard deviation is then generated based on the fitted model. This process is repeated for each subsequent time point, producing a time series of forecasted volatilities. This rolling window approach allows the model parameters to adapt over time and provides forecasts that are responsive to changes in the underlying volatility dynamics.

6.b. Realized Variance Analysis

We used the realized volatility (RV) of our high frequency returns to estimate the quadratic variation of the price processes, as it converges in probability (under ideal conditions) to the quadratic variation. We computed the realized volatility (RV) as the sum of squared log returns over a fixed interval. To identify at which sampling frequency we would achieve the optimal bias-variance trade-off between microstructure noise and statistical precision, we retrieved signature volatility plots for three different time periods for our two time series. We selected three well-spaced years within our sample period to observe how the behaviour of realized volatility has evolved over time and to capture potential structural changes in market microstructure or liquidity conditions. The realized volatility plots were computed by averaging the daily sum of squared log returns over 14 consecutive days. Each selected time period was chosen as a representative window of the corresponding year's return dynamics. To ensure that the plots used were not outliers, we generated volatility signature plots for every two-week window within each year under analysis and selected plots that reflected the typical behaviour observed across those intervals. The results are displayed in Figure 4.



(a) BTC — 2017-08-18 (b) BTC — 2023-11-27 (c) BTC — 2025-03-13

(d) ETH — 2017-08-21 (e) ETH — 2023-09-25 (f) ETH — 2025-03-13

Figure 4: Volatility Signature Plots for BTC (top) and ETH (bottom) across time periods

We observe that panel a and b (2017) show the classical pattern of signature volatility plots. The sample RV decline sharply as the sampling intervals increase, reflecting strong microstructure noise at high frequencies that inflate our volatility estimates. Notably, the plots for the more recent years (2023 and 2025) appear remarkably flat across all sampling intervals. This suggest that Bitcoin and Ethereum have become so liquid in recent years that even a 1-minute sampling frequency is sufficient for constructing reliable RV estimates. We use this as our main motivation and justification for calculating our RV at a 1-minute sampling frequency and using this as a comparison to our volatility forecasts in subsequent sections.

6.c. Visual Evaluation

To get an initial overview of how the volatility forecasts of our four best models compare to our RV volatility estimate, we retrieved scatter plots of each forecast against the RV volatility sequence. These results are displayed in Figure 5 and Figure 6.

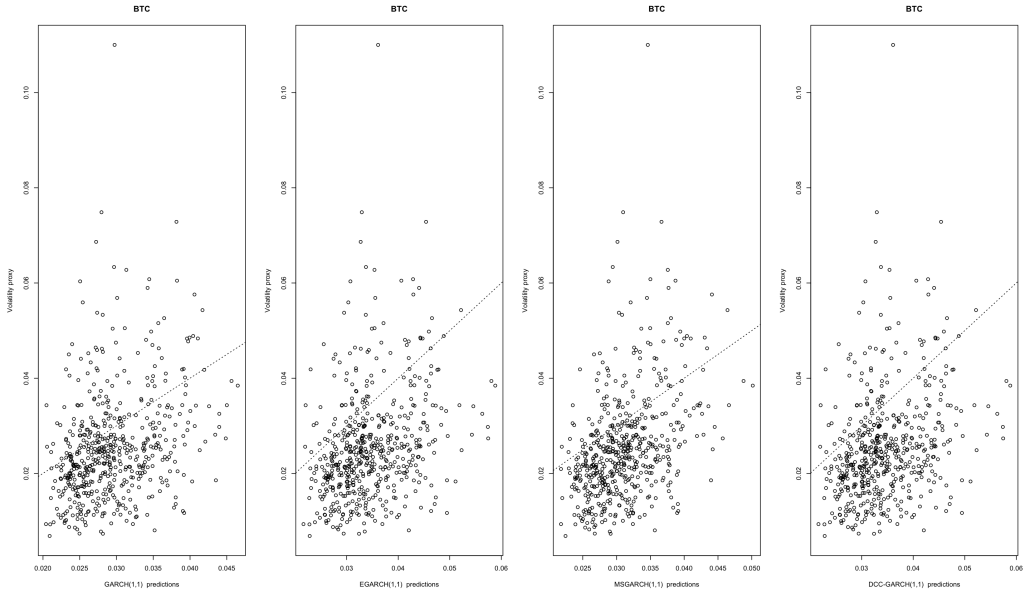


Figure 5: Scatter plot of BTC volatility proxy against predictions from multiple GARCH-type models

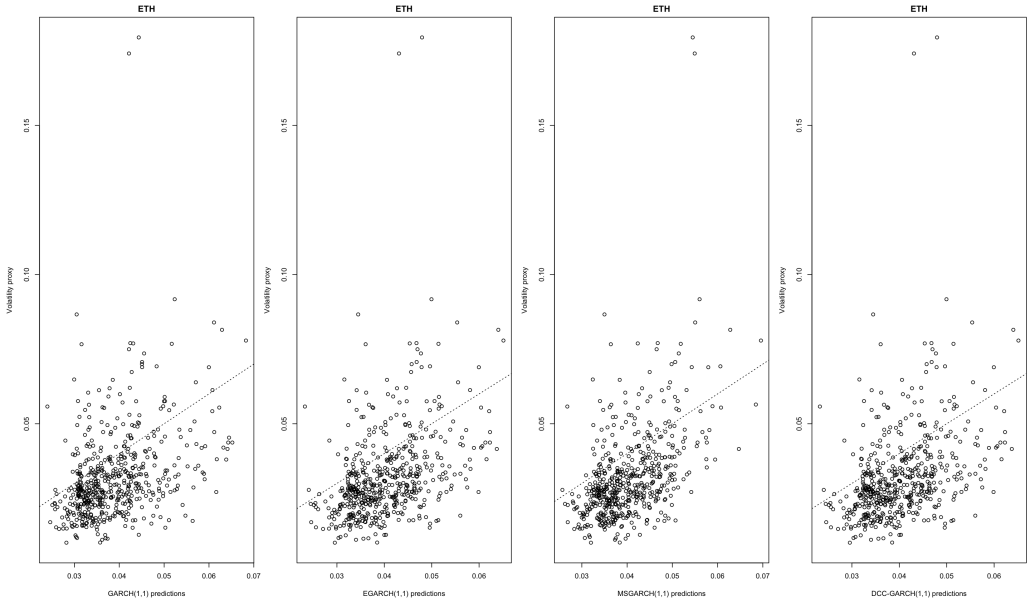


Figure 6: Scatter plot of ETH volatility proxy against predictions from multiple GARCH-type models

For out BTC fitted models, we observe that across all models, most points are clustered in the lower left, which means that that both realized and predicted volatilities are generally low. The points are widely dispersed, particularly for higher predicted values. This suggests that none of the models capture extreme volatility levels very well. We can also observe the the MS-GARCH model exhibits more vertical dispersion, implying to us less precision in the forecasts. The figure of the models predicting ETH volatilities exhibit identical distributions and characteristics. However, the ETH forecasts seem slightly less dispersed, suggesting that the models performed marginally better in predicting volatilities for ETH than for BTC. As expected, the EGARCH models and DCC-GARCH models look almost identical, both for BTC and ETH. Indeed, the differences between the two model forecasts were almost zero. This is

theoretically valid, as our DCC-GARCH was fitted on univariate EGARCH models for both BTC and ETH. Thus, in our upcoming model evaluation, we omit the DCC-GARCH and focus purely on the forecasts of the standard GARCH, EGARCH, and MS-GARCH.

To get an time series overview of how well our four main models where able to forecast, we retrieved time series plots of our volatility forecasts, overlapped with the time series of our realized variance, the results of which are summarized in Figure 7 and Figure 8.

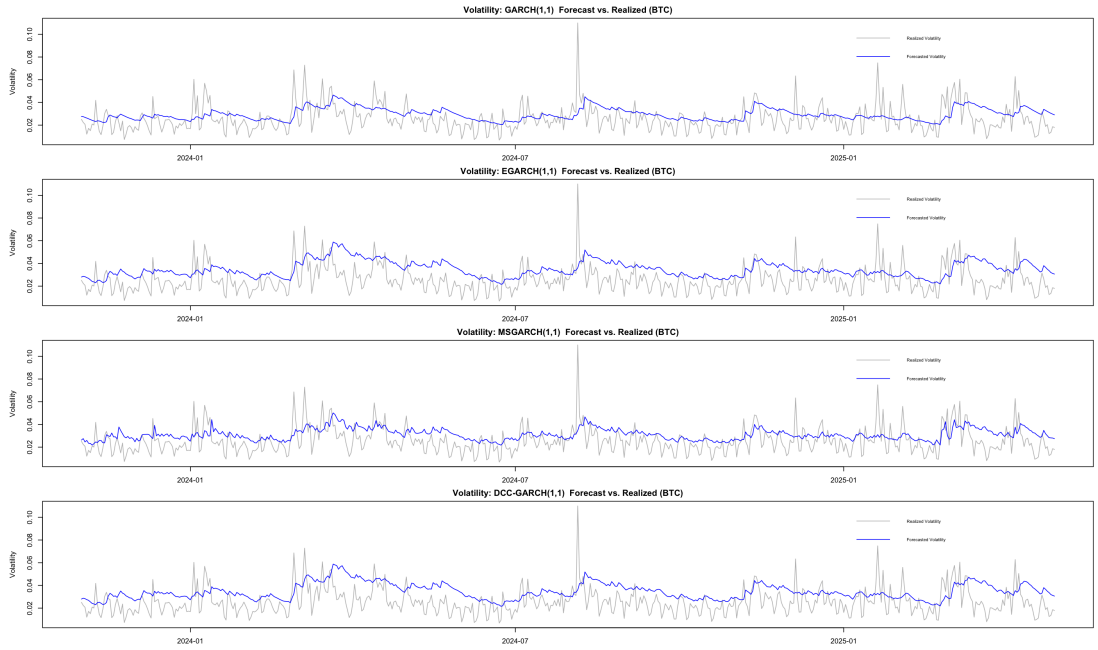


Figure 7: BTC volatility proxy and predicted volatility over time from various GARCH-type models

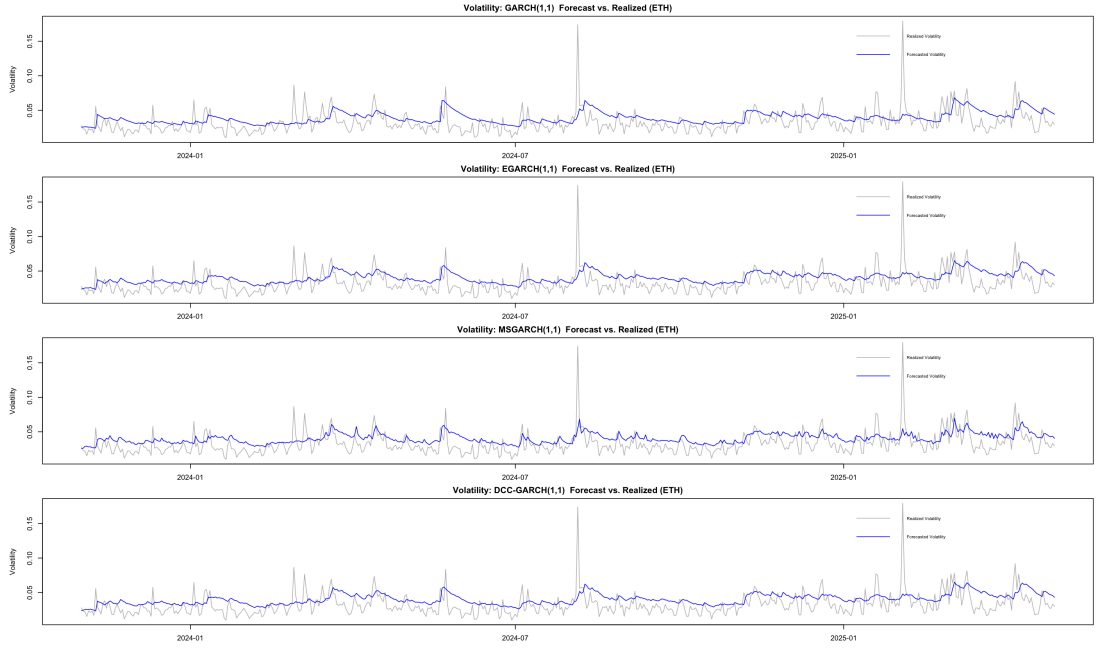


Figure 8: ETH volatility proxy and predicted volatility over time from various GARCH-type models

We observe that across both assets, all models broadly follow the overall trends in realized

volatility, capturing major spikes and shifts. However, predictions are generally smoother and tend to underestimate the magnitude of sharp volatility surges. This is especially the case during turbulent periods. Among the models, EGARCH(1,1) and DCC-GARCH(1,1) appear to track the realized volatility more responsively, particularly during episodes of high market turbulence.

6.d. Mincer-Zarnowitz (MZ) Regression Test

To get an overview of the unbiasedness and efficiency of each model forecast, we implemented individual Mincer-Zarnowitz regressions. Testing whether $\alpha = 0$ and $\beta = 1$ separately (rather than in a joint F-test) allows us to diagnose whether a model suffers from systematic bias or incorrect scaling of forecasted volatility. Further, we used robust (White) standard errors to ensure valid inference in the presence of heteroscedasticity. The results for BTC and ETH are displayed in Table 4 and Table 5.

Table 4: Mincer-Zarnowitz Regression: BTC Forecast Evaluation

Model	α		β	
	Estimate (p-val)	t-stat	Estimate (p-val)	t-stat
GARCH(1,1)	0.0059 (0.0177)	2.38	0.6677 (1e-04)	-3.88
EGARCH(1,1)	0.0085 (1e-04)	3.92	0.4927 (0)	-7.92
MSGARCH(1,1)	-0.0026 (0.3183)	-1.00	0.9066 (0.2909)	-1.06
DCC-GARCH(1,1)	0.0085 (1e-04)	3.92	0.4927 (0)	-7.92

Table 5: Mincer-Zarnowitz Regression: ETH Forecast Evaluation

Model	α		β	
	Estimate (p-val)	t-stat	Estimate (p-val)	t-stat
GARCH(1,1)	0.0059 (0.0398)	2.06	0.6832 (1e-04)	-4.05
EGARCH(1,1)	6e-04 (0.844)	0.20	0.7909 (0.0116)	-2.53
MSGARCH(1,1)	-0.0111 (0.0176)	-2.38	1.0792 (0.5199)	0.64
DCC-GARCH(1,1)	6e-04 (0.844)	0.20	0.7909 (0.0116)	-2.53

We see that for BTC, all models except the MSGARCH(1,1) exhibit a statistically significant intercept, meaning they are biased in their level forecasts. GARCH(1,1) and EGARCH(1,1) both have $\alpha > 0$ indicating they systematically overpredict volatility. Their $\beta > 0$ coefficients are also significantly different from 1, and specifically less than 1, suggesting they underreact to actual movements in volatility. DCC-GARCH(1,1) mirrors EGARCH(1,1), showing both bias and inefficiency. MSGARCH(1,1), however, does not significantly reject either $\alpha = 0$ or $\beta = 0$, suggesting it is neither significantly biased nor inefficient.

For ETH, the we can observe more mixed results. GARCH(1,1) again shows both a significant $\alpha > 0$ and $\beta < 1$, pointing to upward bias and underreaction, as with BTC. EGARCH(1,1) and DCC-GARCH(1,1) do not show significant bias, but both reject $\beta = 0$ with $\beta < 1$, meaning they still underreact to volatility changes. MSGARCH(1,1) is the only model with a significantly negative α, indicating downward bias, and its β is not significantly different from 1, which suggests potential correct scaling but poor calibration. Overall, the MZ regressions show that none of the models fully satisfy both MZ efficiency conditions, which casts doubt on the reliability of volatility forecasts

6.e. Superior Predictive Ability (SPA) Analysis

To evaluate whether any of our GARCH-type models significantly outperformed the others in forecasting volatility, we conducted the Superior Predictive Ability (SPA) test. In our implementation, we iteratively set each model as the benchmark once. This ensured that performance was evaluated symmetrically across all specifications. The comparison was conducted using both the QLIKE and MSE loss functions to capture different aspects of forecast accuracy. The results are displayed in Tables 6 and Tables 7.

Table 6: SPA Test Results for BTC Volatility Forecasts

Benchmark Model	**QLIKE (p-value)**	**MSE (p-value)**
GARCH(1,1)	4.4135 (0.0000)	0.000138 (0.7356)
EGARCH(1,1)	4.1318 (0.7422)	0.000214 (0.0000)
MSGARCH(1,1)	4.3020 (0.0000)	0.000143 (0.0902)

Table 7: SPA Test Results for ETH Volatility Forecasts

Benchmark Model	**QLIKE (p-value)**	**MSE (p-value)**
GARCH(1,1)	4.1091 (0.0000)	0.000273 (0.2493)
EGARCH(1,1)	4.0360 (0.0920)	0.000283 (0.0241)
MSGARCH(1,1)	4.0223 (0.7520)	0.000261 (0.7611)

For BTC, the SPA results based on the QLIKE loss function indicate that both the GARCH(1,1) and MSGARCH(1,1) models are significantly outperformed by at least one competing model, as reflected in their p-values of 0.0000. In contrast, the EGARCH(1,1) model remains within the set of superior models (p-value = 0.7422), suggesting it offers more robust volatility forecasts under this loss. When evaluating performance under the MSE loss, the results are different. EGARCH(1,1) is now significantly outperformed (p-value = 0.0000), while both GARCH(1,1) and MSGARCH(1,1) show no significant inferiority, with p-values of 0.7356 and 0.0902, respectively. This highlights how conclusions about model performance can vary depending on the selected loss function.

For ETH, under the QLIKE loss, GARCH(1,1) is clearly outperformed (p-value = 0.0000), whereas both EGARCH(1,1) and MSGARCH(1,1) remain within the set of superior models with p-values of 0.0920 and 0.7520, respectively. Under the MSE loss, however, no model is found to be significantly outperformed, with all p-values exceeding conventional significance levels. These results show us that model evaluation is sensitive to the chosen loss function and we need underscore our motivation of the chosen loss function with economic reasoning.

6.f. Model Confidence Set (MCF) Test

To further investigate the relative performance of our forecasting models, we implemented the Model Confidence Set (MCS) test. This iterative testing framework identifies the subset of models that cannot be statistically distinguished from the best-performing model at a given confidence level. We applied the MCS procedure using both the QLIKE and MSE loss functions and evaluated performance at the 5% and 10% significance levels.

We compared the models by looking at differences in their forecast errors and used bootstrapping to test whether any model performed significantly worse than the others. At each step, the

worst model was removed until only models with similar predictive ability remained. The final sets of best-performing models for BTC and ETH are shown in the following Tables 8 and Tables 9.

Table 8: Model Confidence Set (MCS) Results for BTC Volatility Forecasts

Loss Function	**Significance Level**	**Models in MCS**	**Eliminated Models**
QLIKE	5%	EGARCH(1,1)	GARCH(1,1), MSGARCH(1,1)
QLIKE	10%	EGARCH(1,1)	GARCH(1,1), MSGARCH(1,1)
MSE	5%	GARCH(1,1), MSGARCH(1,1)	EGARCH(1,1)
MSE	10%	GARCH(1,1)	EGARCH(1,1), MSGARCH(1,1)

Table 9: Model Confidence Set (MCS) Results for ETH Volatility Forecasts

Loss Function	**Significance Level**	**Models in MCS**	**Eliminated Models**
QLIKE	5%	MSGARCH(1,1)	GARCH(1,1), EGARCH(1,1)
QLIKE	10%	MSGARCH(1,1)	GARCH(1,1), EGARCH(1,1)
MSE	5%	GARCH(1,1), MSGARCH(1,1)	EGARCH(1,1)
MSE	10%	GARCH(1,1), MSGARCH(1,1)	EGARCH(1,1)

We observe that for BTC, the EGARCH(1,1) model consistently remained in the MCS under the QLIKE loss function, suggesting it had the most robust performance in terms of capturing the distributional characteristics of the realized volatility. Under the MSE loss function, both GARCH(1,1) and MSGARCH(1,1) models were retained at the 5% level, but only GARCH(1,1) remained at the 10% level, indicating that in terms of squared error, GARCH(1,1) was more stable over broader significance levels.

For ETH, the MSGARCH(1,1) model dominated under the QLIKE loss function, being the only model retained at both significance levels. This indicates superior performance in modeling the distribution of volatility. Under the MSE criterion, both GARCH(1,1) and MSGARCH(1,1) remained in the MCS at both significance levels, showing that these models performed comparably well in minimizing forecast error magnitudes.

7 Conclusion

This paper investigates the forecasting performance of several GARCH-type volatility models applied to Bitcoin and Ethereum return series, focusing on both in-sample model fit and out-of-sample predictive accuracy. The analysis confirms that cryptocurrency return volatility exhibits strong clustering, heavy tails, and limited leverage effects, thereby validating the use of AR(1)-GARCH-type processes with Student-t innovations.

The in-sample evaluation shows that AR(1)-GARCH(1,1) and AR(1)-EGARCH(1,1) models consistently offer strong fit across daily and higher-frequency return series. The EGARCH specification captures asymmetric volatility responses and exhibits significant persistence and kurtosis-adjusted dynamics. Diagnostic tests on residuals and squared residuals confirm the appropriateness of these specifications, with the EGARCH(1,1) model particularly successful at filtering conditional heteroskedasticity. For multivariate modeling, the DCC-GARCH(1,1) formulation provides meaningful insights into time-varying return correlations and co-movement patterns, especially during volatile episodes.

Out-of-sample analysis, however, presents a more complex picture. While GARCH(1,1) and EGARCH(1,1) produce smooth volatility forecasts that broadly track realized variance, they exhibit underreaction during abrupt market shifts. This is confirmed by Mincer-Zarnowitz regression results, which indicate statistically significant deviations from forecast unbiasedness and efficiency. For BTC, both EGARCH(1,1) and DCC-GARCH(1,1) overpredict volatility on average and fail to fully adjust to changes in realized variance, with slope coefficients significantly below one. MSGARCH(1,1), while less biased, displays greater dispersion and less reliable calibration. For ETH, forecast bias varies across models, with MSGARCH(1,1) uniquely showing a significant negative intercept, suggesting consistent underestimation.

Forecast performance further depends on the chosen evaluation criterion. Under the QLIKE loss function, which is robust to measurement error in the volatility proxy, the Model Confidence Set (MCS) test at the 5% significance level identifies EGARCH(1,1) as the only superior model for BTC. For ETH, however, the MSGARCH(1,1) model remains in the confidence set under QLIKE. In contrast, the MSE loss function favors the GARCH(1,1) and MSGARCH(1,1) models for both assets. These results underline the sensitivity of model evaluation to the selected loss function and highlight the importance of aligning performance metrics with the intended forecasting application.

Overall, no single model consistently dominates across all metrics. The EGARCH(1,1) specification offers the most reliable performance under distributionally robust loss functions, confirming its practical relevance for capturing nonlinear volatility features. Nonetheless, its relative performance deteriorates under squared-error loss. The DCC-GARCH model enhances the interpretability of dynamic correlation structures but does not offer meaningful out-of-sample forecast improvements compared to its univariate margins. The findings suggest that model selection for cryptocurrency volatility forecasting must account for both statistical performance and robustness to extreme market conditions. Future extensions could incorporate long-memory or stochastic volatility frameworks, or hybrid models integrating high-frequency information to address residual inefficiencies and improve responsiveness to sudden regime shifts.

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Appendix

Supplementary Tables

Parameter	Estimate	Std. Error	t value	Pr(>|t|)
μ	0.000855	0.000512	1.669	0.0951
AR(1)	-0.058801	0.019049	-3.087	0.0020
ω	0.000009	0.000031	0.300	0.7641
α_1	0.064888	0.019714	3.291	0.0010
β_1	0.934112	0.047788	19.547	< 0.0001
Shape	3.218258	0.368167	8.741	< 0.0001
Model Fit				
Log-Likelihood (ℓ)	4570.321			
Number of Parameters (k)	6			
AIC	-9134.642			
BIC	-9094.287			
Diagnostic p-values				
McLeod-Li (squared residuals)	0.948			
Ljung-Box (standard residuals)	0.064			

Table A1: BTC Daily S-GARCH ARMA-GARCH(1,0;1,1) In-Sample Estimation and Diagnostics (Robust Standard Errors)

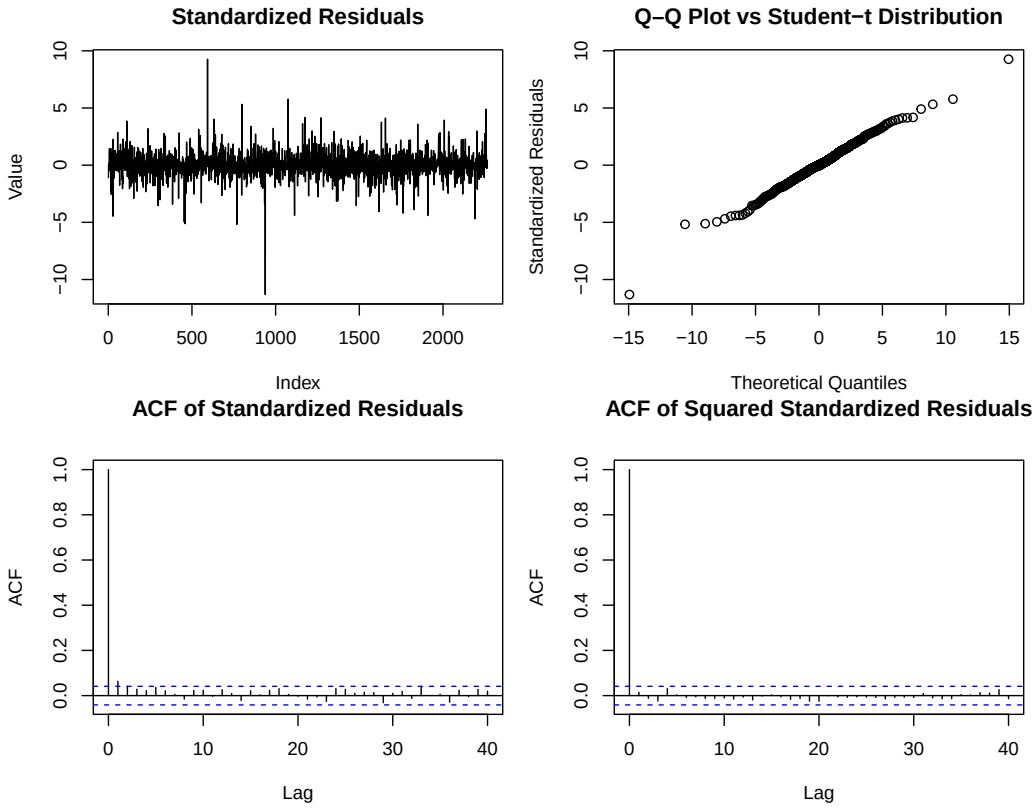


Figure A1: Residual Diagnostics based on z_t for BTC Daily S-GARCH ARMA-GARCH(1,0;1,1)

Parameter	Estimate	Std. Error	t value	Pr(>|t|)
μ	0.000282	0.000086	3.288	0.0010
AR(1)	-0.037923	0.009478	-4.001	0.0001
ω	0.000001	0.000010	0.096	0.9234
α_1	0.072001	0.084510	0.852	0.3942
β_1	0.540964	0.049993	10.821	< 0.0001
β_2	0.386035	0.040918	9.434	< 0.0001
Shape	3.066460	0.081575	37.591	< 0.0001
Model Fit				
Log-Likelihood (ℓ)	25597.23			
Number of Parameters (k)	7			
AIC	-51187.47			
BIC	-51130.69			
Diagnostic p-values				
McLeod-Li (squared residuals)	0.633			
Ljung-Box (standard residuals)	6.88×10^{-15}			

Table A2: BTC 6-Hour S-GARCH ARMA-GARCH(1,0;1,2) In-Sample Estimation and Diagnostics (Robust Standard Errors)

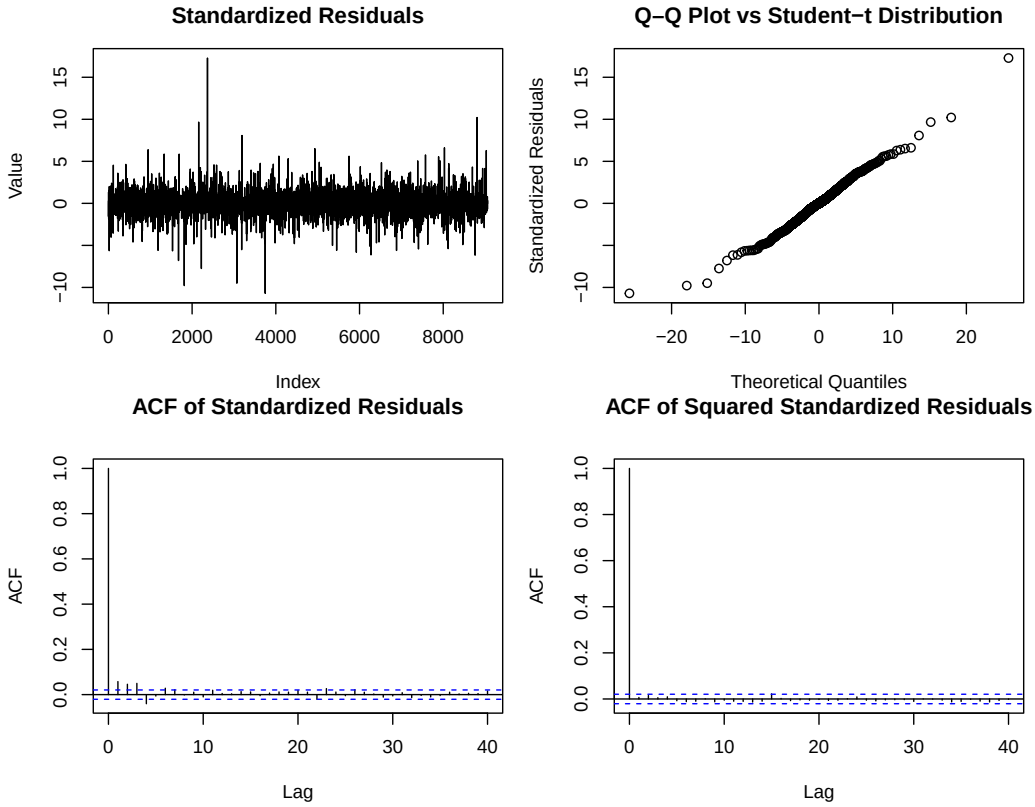


Figure A2: Residual Diagnostics based on z_t for BTC 6-Hour S-GARCH ARMA-GARCH(1,0;1,2)

Parameter	Estimate	Std. Error	t value	Pr(>|t|)
μ	0.000981	0.000711	1.379	0.16789
AR(1)	-0.096196	0.018922	-5.084	< 0.0001
ω	0.000031	0.000054	0.575	0.56518
α_1	0.089983	0.032446	2.773	0.00555
β_1	0.909017	0.052283	17.387	< 0.0001
Shape	3.487737	0.262347	13.294	< 0.0001
Model Fit				
Log-Likelihood (ℓ)	3955.021			
Number of Parameters (k)	6			
AIC	-7904.042			
BIC	-7863.687			
Diagnostic p-values				
McLeod-Li (squared residuals)	0.69016			
Ljung-Box (standard residuals)	0.00074			

Table A3: ETH Daily S-GARCH ARMA-GARCH(1,0;1,1) In-Sample Estimation and Diagnostics (Robust Standard Errors)

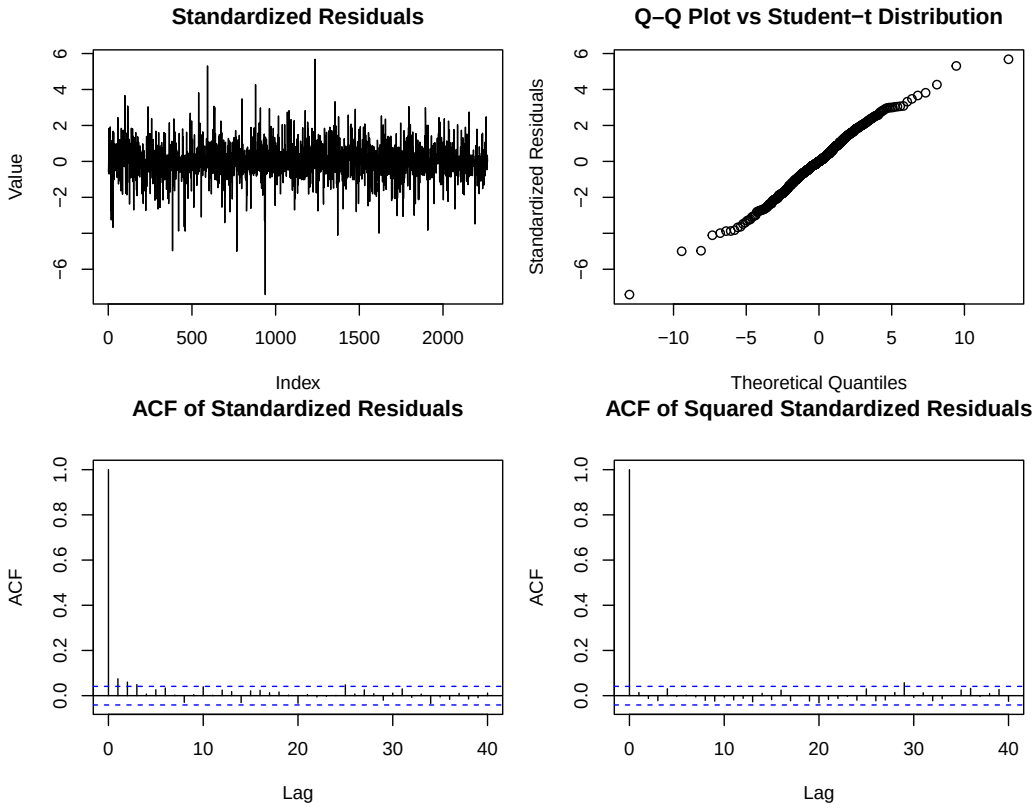


Figure A3: Residual Diagnostics based on z_t for ETH Daily S-GARCH ARMA-GARCH(1,0;1,1)

Parameter	Estimate	Std. Error	t value	Pr(>|t|)
μ	0.000232	0.000163	1.424	0.1544
AR(1)	-0.037156	0.027633	-1.345	0.1788
ω	0.000003	0.000148	0.020	0.9840
α_1	0.080312	0.609329	0.132	0.8951
β_1	0.918688	0.750712	1.224	0.2210
Shape	3.427322	0.808763	4.238	< 0.0001
Model Fit				
Log-Likelihood (ℓ)	23118.26			
Number of Parameters (k)	6			
AIC	-46230.52			
BIC	-46181.85			
Diagnostic p-values				
McLeod-Li (squared residuals)	0.057			
Ljung-Box (standard residuals)	0.000			

Table A4: ETH 6-Hour S-GARCH ARMA-GARCH(1,0;1,1) In-Sample Estimation and Diagnostics (Robust Standard Errors)

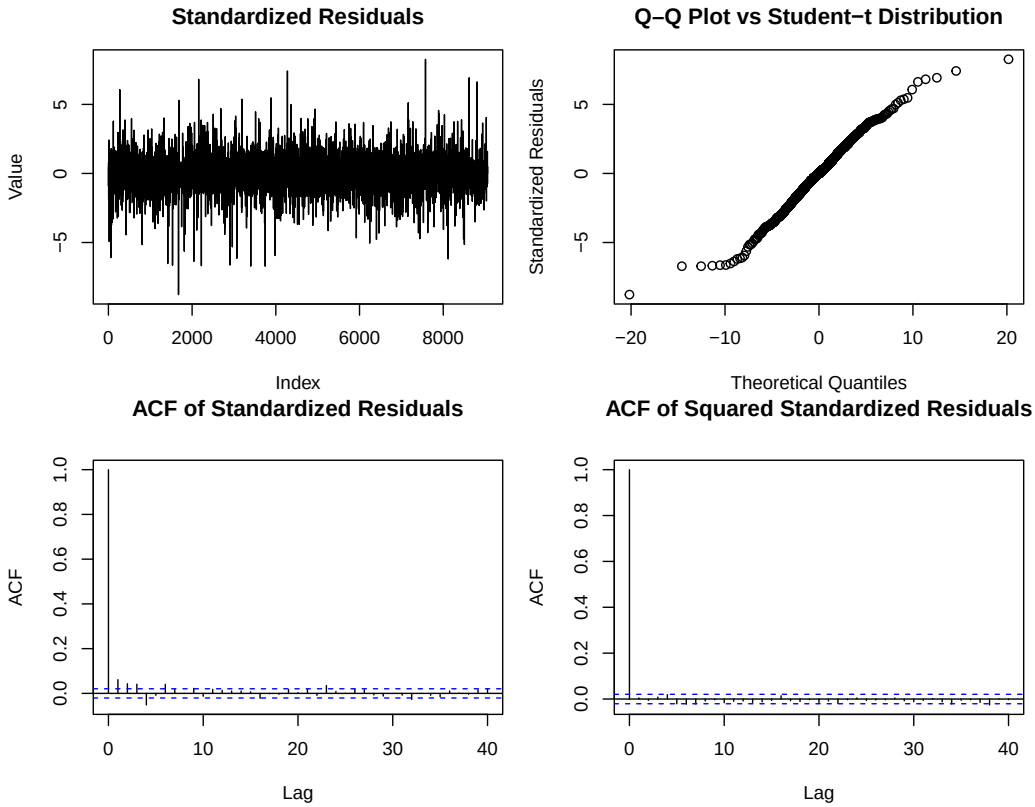


Figure A4: Residual Diagnostics based on z_t for ETH 6-Hour S-GARCH ARMA-GARCH(1,0;1,1)

Parameter	Estimate	Std. Error	t value	Pr(>|t|)
μ	0.000720	0.000516	1.397	0.1625
AR(1)	-0.064310	0.017445	-3.686	0.0002
ω	0.000220	0.000197	1.113	0.2659
α_1	0.112386	0.023752	4.732	< 0.0001
β_1	0.931702	0.013274	70.192	< 0.0001
$\eta_{1,1}$	-0.023898	0.094875	-0.252	0.8011
Shape	2.692023	0.174618	15.417	< 0.0001
Model Fit				
Log-Likelihood (ℓ)	4583.5			
Number of Parameters (k)	7			
AIC	-9160.001			
BIC	-9112.92			
Diagnostic p-values				
McLeod-Li (squared residuals)	0.982			
Ljung-Box (standard residuals)	0.017			

Table A5: BTC Daily T-GARCH ARMA-GARCH(1,0;1,1) In-Sample Estimation and Diagnostics (Robust Standard Errors)

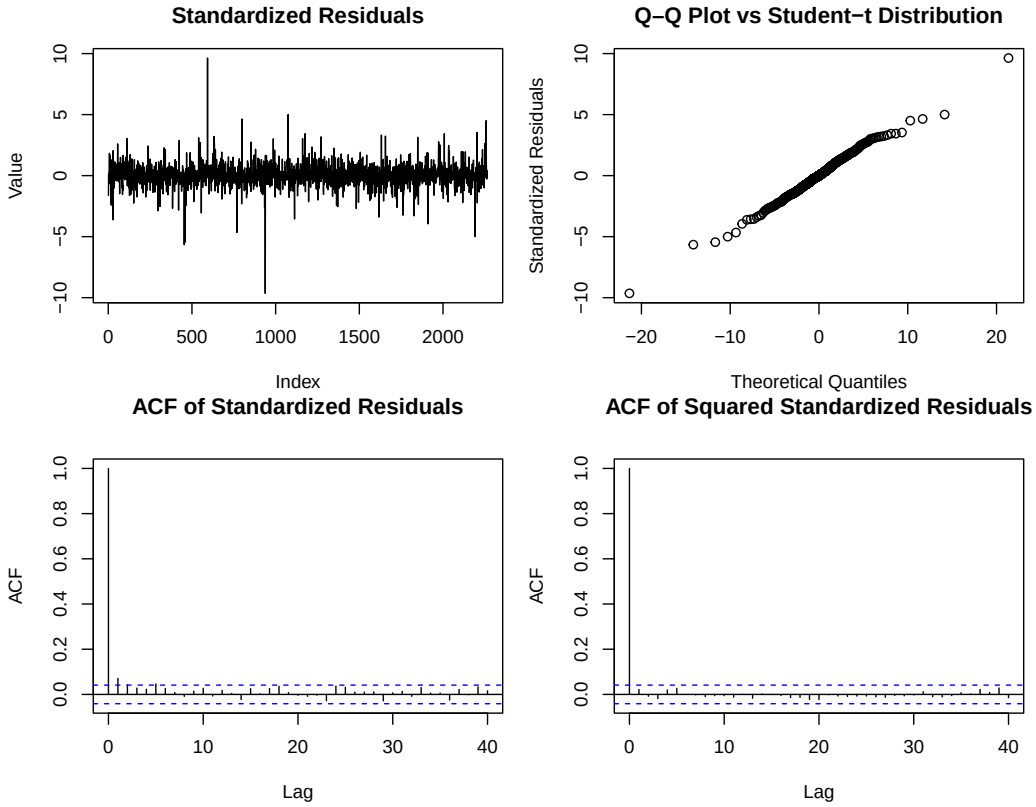


Figure A5: Residual Diagnostics based on z_t for BTC Daily T-GARCH ARMA-GARCH(1,0;1,1)

Parameter	Estimate	Std. Error	t value	Pr(>|t|)
μ	0.000216	0.000082	2.622	0.0087
MA(1)	-0.040568	0.007908	-5.130	< 0.0001
ω	0.000084	0.000173	0.486	0.6271
α_1	0.121875	0.069405	1.756	0.0791
β_1	0.936984	0.045303	20.682	< 0.0001
$\eta_{1,1}$	0.064431	0.051354	1.255	0.2096
Shape	2.411249	0.085813	28.099	< 0.0001
Model Fit				
Log-Likelihood (ℓ)	25689.42			
Number of Parameters (k)	7			
AIC	-51371.83			
BIC	-51315.05			
Diagnostic p-values				
McLeod-Li (squared residuals)	0.785			
Ljung-Box (standard residuals)	0.000			

Table A6: BTC 6-Hour T-GARCH ARMA-GARCH(0,1;1,1) In-Sample Estimation and Diagnostics (Robust Standard Errors)

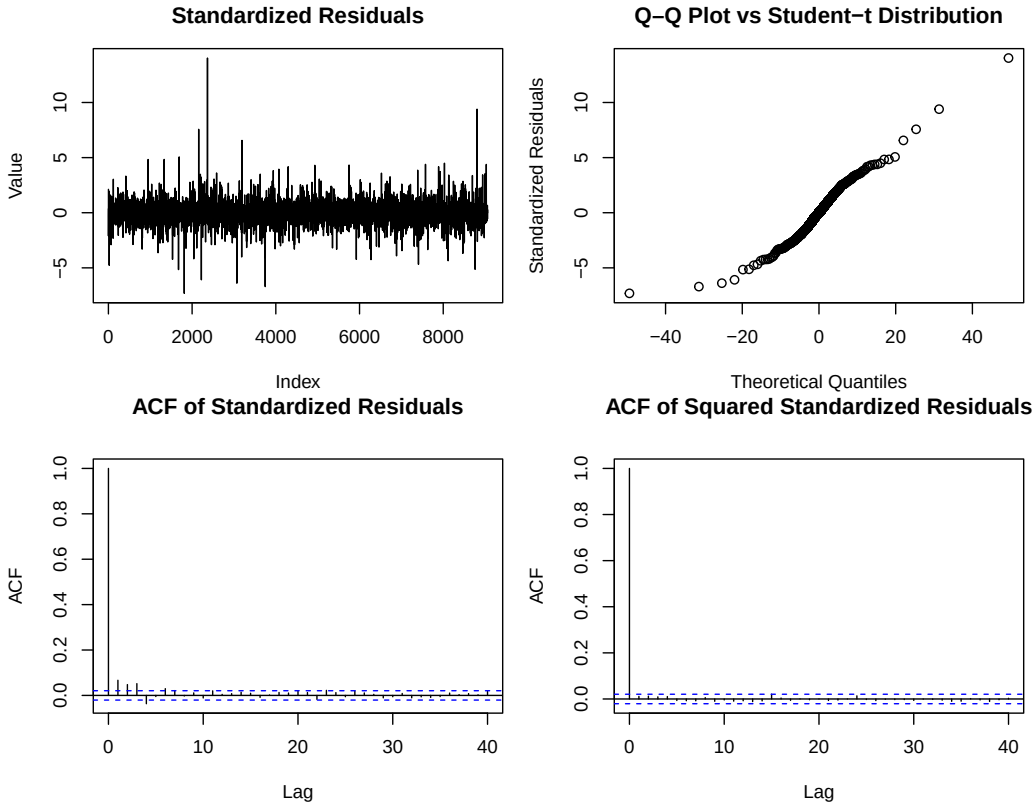


Figure A6: Residual Diagnostics based on z_t for BTC 6-Hour T-GARCH ARMA-GARCH(0,1;1,1)

Parameter	Estimate	Std. Error	t value	Pr(>|t|)
μ	0.000755	0.000702	1.076	0.2821
AR(1)	-0.096747	0.018565	-5.211	< 0.0001
ω	0.000508	0.000476	1.068	0.2856
α_1	0.104788	0.034752	3.015	0.0026
β_1	0.923477	0.028210	32.736	< 0.0001
$\eta_{1,1}$	-0.017748	0.094260	-0.188	0.8507
Shape	3.169658	0.253644	12.496	< 0.0001
Model Fit				
Log-Likelihood (ℓ)	3963.92			
Number of Parameters (k)	7			
AIC	-7920.84			
BIC	-7873.76			
Diagnostic p-values				
McLeod-Li (squared residuals)	0.505			
Ljung-Box (standard residuals)	0.0017			

Table A7: ETH Daily T-GARCH ARMA-GARCH(1,0;1,1) In-Sample Estimation and Diagnostics (Robust Standard Errors)

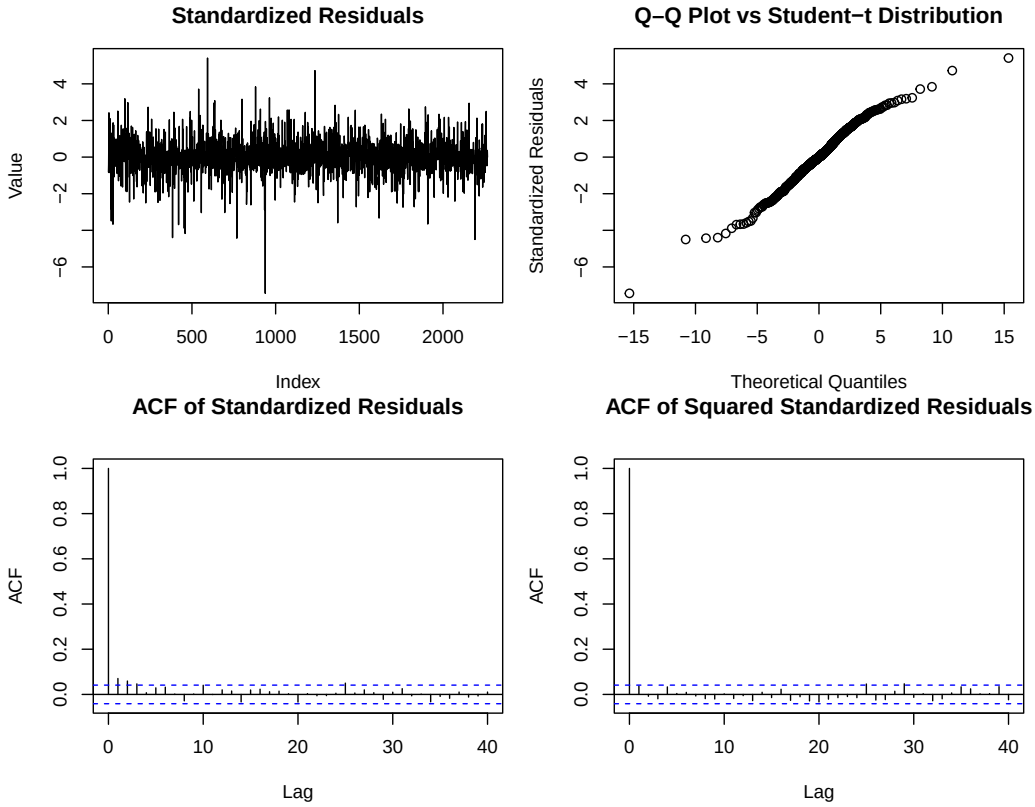


Figure A7: Residual Diagnostics based on z_t for ETH Daily T-GARCH ARMA-GARCH(1,0;1,1)

Parameter	Estimate	Std. Error	t value	Pr(>|t|)
μ	0.000137	0.000120	1.142	0.2536
AR(1)	-0.036923	0.008726	-4.232	< 0.0001
ω	0.000162	0.000102	1.592	0.1114
α_1	0.132324	0.025958	5.098	< 0.0001
β_1	0.916630	0.018018	50.873	< 0.0001
$\eta_{1,1}$	0.117888	0.045820	2.573	0.0101
Shape	2.806409	0.098530	28.483	< 0.0001
Model Fit				
Log-Likelihood (ℓ)	23183.83			
Number of Parameters (k)	7			
AIC	-46360.66			
BIC	-46303.88			
Diagnostic p-values				
McLeod-Li (squared residuals)	0.0159			
Ljung-Box (standard residuals)	0.000			

Table A8: ETH 6-Hour T-GARCH ARMA-GARCH(1,0;1,1) In-Sample Estimation and Diagnostics (Robust Standard Errors)

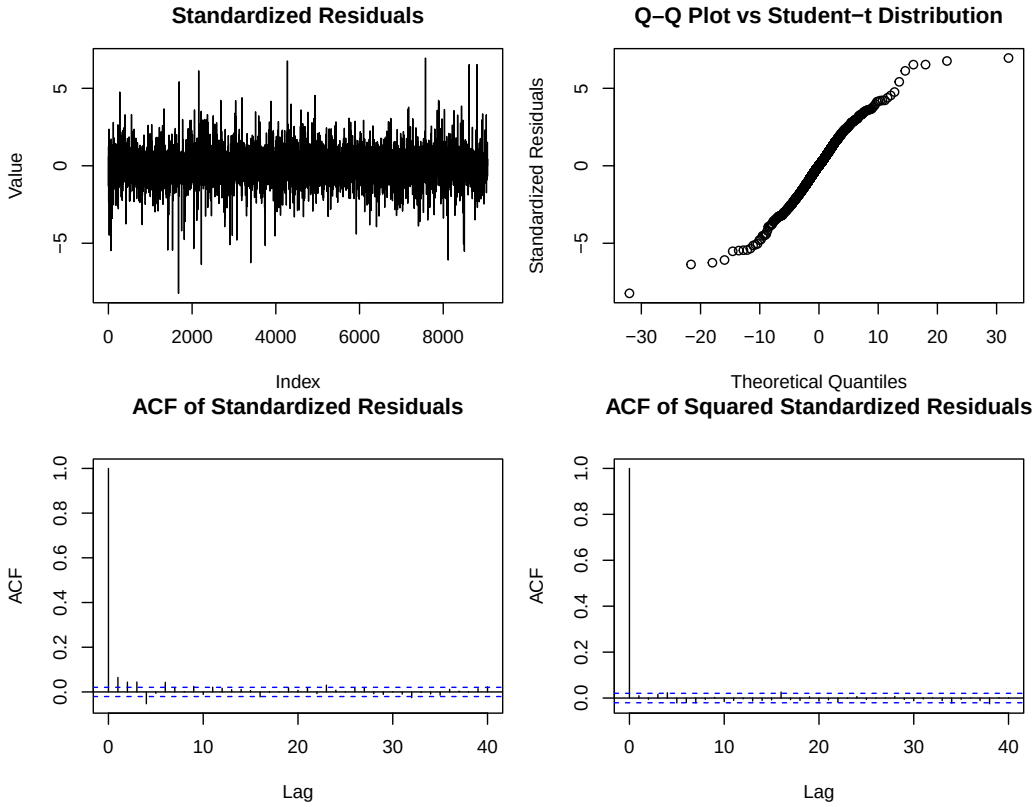


Figure A8: Residual Diagnostics based on z_t for ETH 6-Hour T-GARCH ARMA-GARCH(1,0;1,1)

Parameter	Estimate	Std. Error	t value	Pr(>|t|)
μ	0.000863	0.000533	1.620	0.1052
AR(1)	-0.059210	0.019950	-2.968	0.0030
ω	0.000009	0.000056	0.156	0.8759
α_1	0.066922	0.021820	3.067	0.0022
β_1	0.935758	0.088097	10.622	< 0.0001
γ_1	-0.007360	0.042121	-0.175	0.8613
Shape	3.219610	0.600286	5.363	< 0.0001
Model Fit				
Log-Likelihood (ℓ)	4570.454			
Number of Parameters (k)	7			
AIC	-9133.907			
BIC	-9086.827			
Diagnostic p-values				
McLeod-Li (squared residuals)	0.953			
Ljung-Box (standard residuals)	0.070			

Table A9: BTC Daily GJR-GARCH ARMA-GARCH(1,0;1,1) In-Sample Estimation and Diagnostics (Robust Standard Errors)

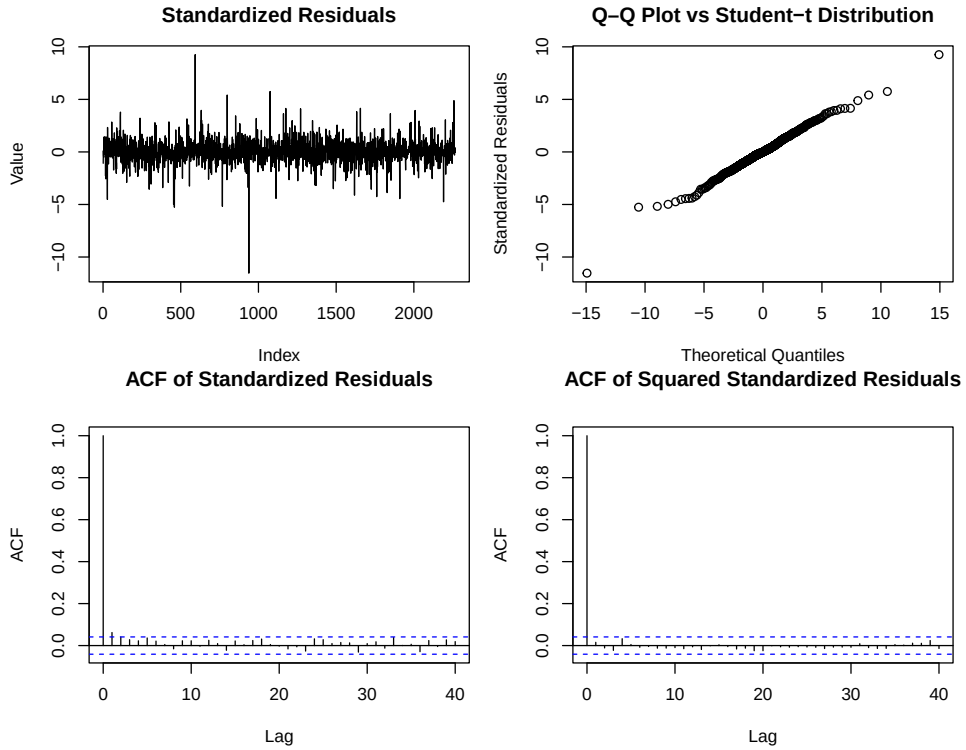


Figure A9: Residual Diagnostics based on z_t for BTC Daily GJR-GARCH ARMA-GARCH(1,0;1,1)

Parameter	Estimate	Std. Error	t value	Pr(>|t|)
μ	0.000275	0.000103	2.662	0.0078
AR(1)	-0.038217	0.016728	-2.285	0.0223
ω	0.000001	0.000029	0.023	0.9814
α_1	0.042221	0.030446	1.387	0.1655
α_2	0.000000	0.174071	0.000	1.0000
β_1	0.951105	0.163396	5.821	< 0.0001
γ_1	0.098248	0.114411	0.859	0.3905
γ_2	-0.086900	0.149225	-0.582	0.5603
Shape	3.083754	0.906429	3.402	0.0007
Model Fit				
Log-Likelihood (ℓ)	25604.19			
Number of Parameters (k)	9			
AIC	-51199.38			
BIC	-51126.38			
Diagnostic p-values				
McLeod-Li (squared residuals)	0.738			
Ljung-Box (standard residuals)	6.99×10^{-15}			

Table A10: BTC 6-Hour GJR-GARCH ARMA-GARCH(1,0;2,1) In-Sample Estimation and Diagnostics (Robust Standard Errors)

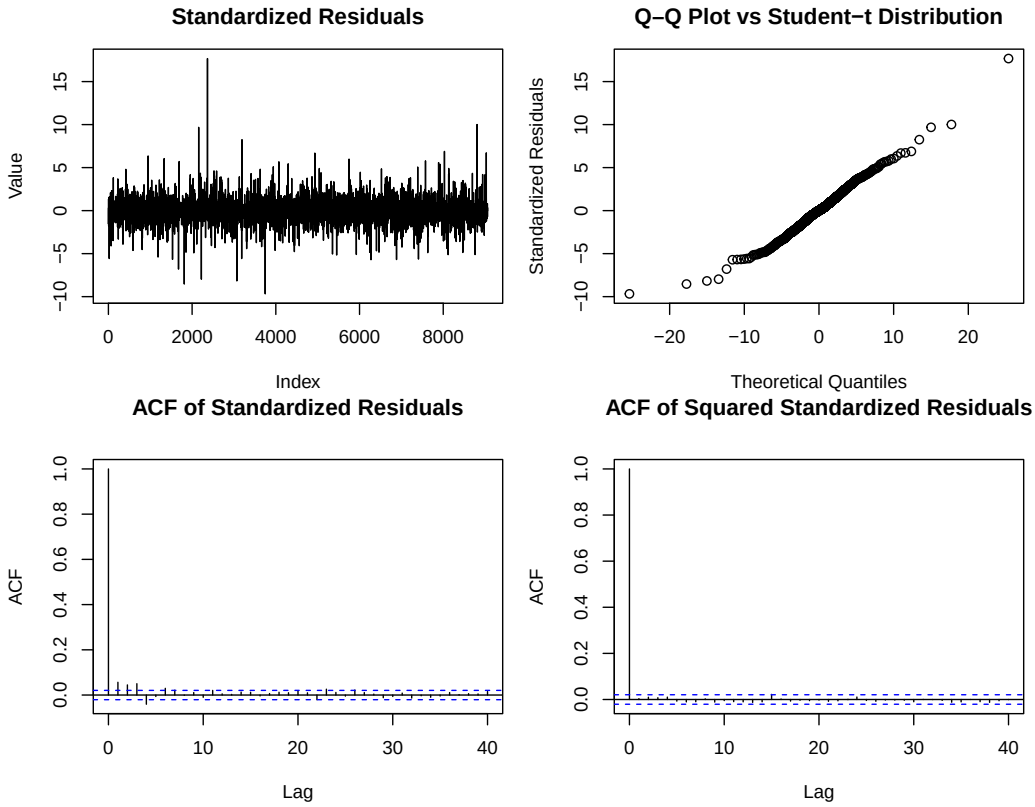


Figure A10: Residual Diagnostics based on z_t for BTC 6-Hour GJR-GARCH ARMA-GARCH(1,0;2,1)

Parameter	Estimate	Std. Error	t value	Pr(>|t|)
μ	0.000995	0.000707	1.407	0.1594
AR(1)	-0.096708	0.019757	-4.895	< 0.0001
ω	0.000029	0.000062	0.471	0.6378
α_1	0.091154	0.031808	2.866	0.0042
β_1	0.911007	0.064159	14.199	< 0.0001
γ_1	-0.006322	0.028406	-0.223	0.8239
Shape	3.494445	0.278044	12.568	< 0.0001
Model Fit				
Log-Likelihood (ℓ)	3955.076			
Number of Parameters (k)	7			
AIC	-7903.152			
BIC	-7856.072			
Diagnostic p-values				
McLeod-Li (squared residuals)	0.684			
Ljung-Box (standard residuals)	0.0007			

Table A11: ETH Daily GJR-GARCH ARMA-GARCH(1,0;1,1) In-Sample Estimation and Diagnostics (Robust Standard Errors)

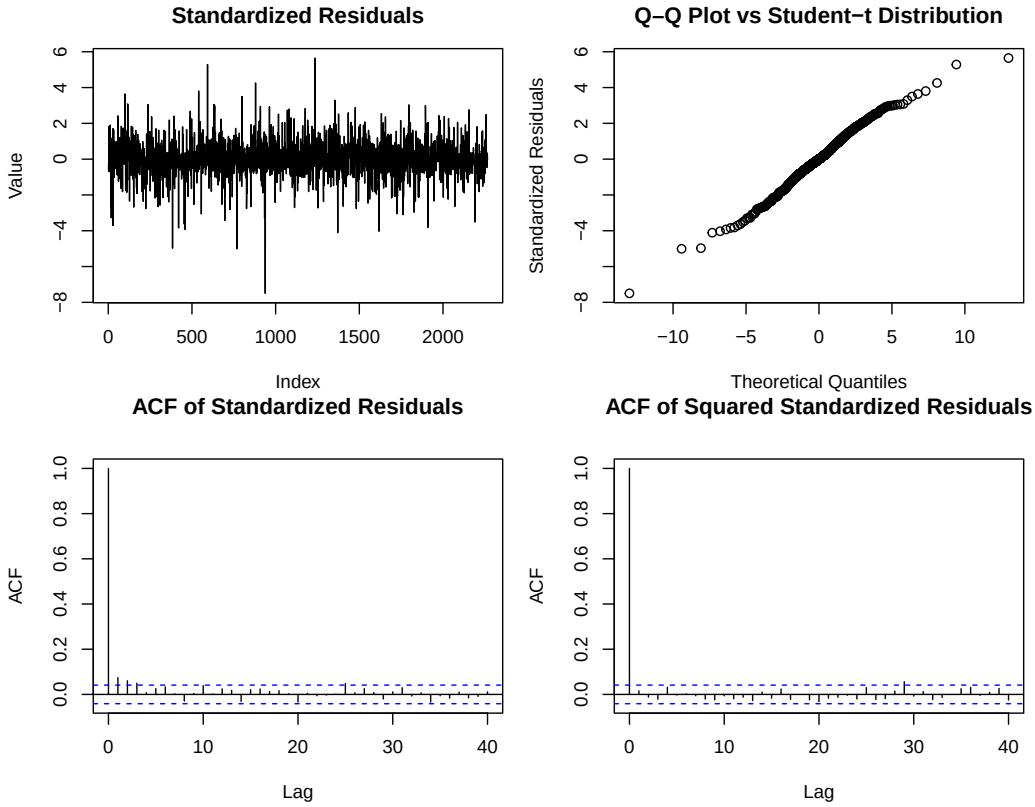


Figure A11: Residual Diagnostics based on z_t for ETH Daily GJR-GARCH ARMA-GARCH(1,0;1,1)

Parameter	Estimate	Std. Error	t value	Pr(>|t|)
μ	0.000204	0.000116	1.757	0.0789
AR(1)	-0.037820	0.011059	-3.420	0.0006
ω	0.000003	0.000002	1.035	0.3006
α_1	0.064831	0.078835	0.822	0.4109
α_2	0.000019	0.093130	0.000	0.9998
β_1	0.926806	0.017626	52.581	< 0.0001
β_2	0.000003	0.006815	0.000	0.9996
γ_1	0.103053	0.064858	1.589	0.1121
γ_2	-0.088372	0.063798	-1.385	0.1660
Shape	3.435679	0.119068	28.855	< 0.0001
Model Fit				
Log-Likelihood (ℓ)	23128.93			
Number of Parameters (k)	10			
AIC	-46247.85			
BIC	-46166.74			
Diagnostic p-values				
McLeod-Li (squared residuals)	0.088			
Ljung-Box (standard residuals)	0.000			

Table A12: ETH 6-Hour GJR-GARCH ARMA-GARCH(1,0;2,2) In-Sample Estimation and Diagnostics (Robust Standard Errors)

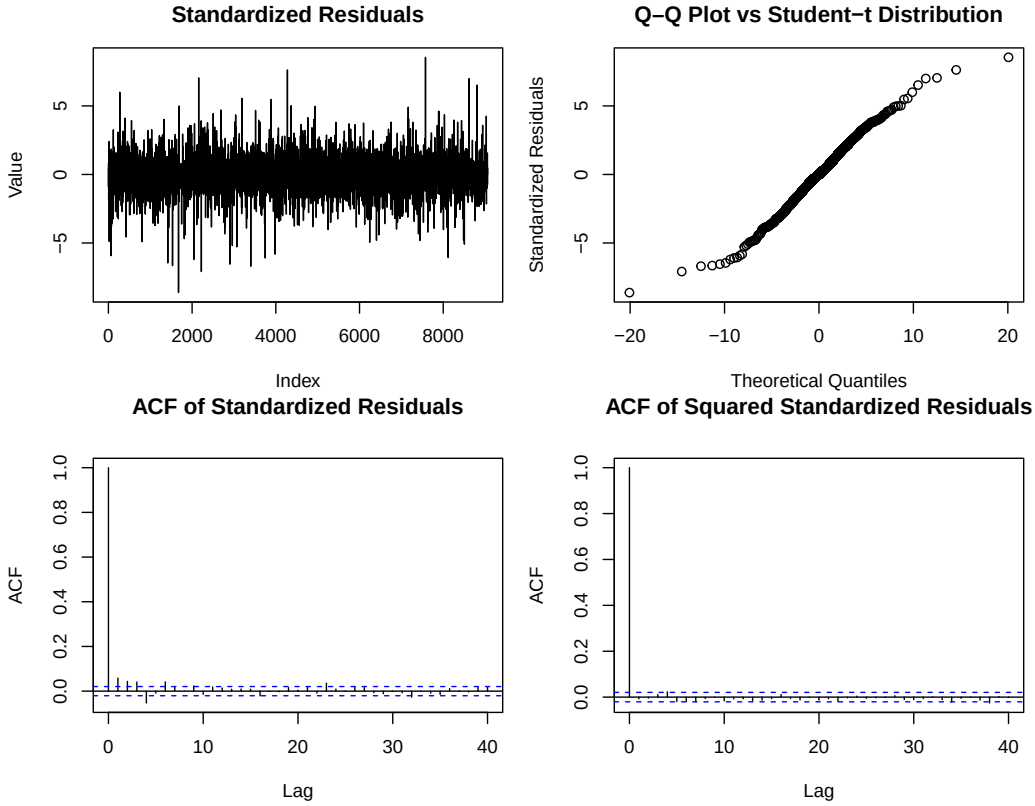


Figure A12: Residual Diagnostics based on z_t for ETH 6-Hour GJR-GARCH ARMA-GARCH(1,0;2,2)

Parameter	Estimate	Std. Error	t value	Pr(>|t|)
μ	0.000723	0.000431	1.680	0.0930
AR(1)	-0.064439	0.016493	-3.907	< 0.0001
ω	-0.018521	0.008366	-2.214	0.0268
α_1	0.005467	0.014014	0.390	0.6964
β_1	0.996841	0.001303	765.124	< 0.0001
γ_1	0.196923	0.013739	14.333	< 0.0001
Shape	2.664547	0.154612	17.234	< 0.0001
Model Fit				
Log-Likelihood (ℓ)	4584.893			
Number of Parameters (k)	7			
AIC	-9162.787			
BIC	-9115.706			
Diagnostic p-values				
McLeod-Li (squared residuals)	0.985			
Ljung-Box (standard residuals)	0.0146			

Table A13: BTC Daily EGARCH ARMA-GARCH(1,0;1,1) In-Sample Estimation and Diagnostics (Robust Standard Errors)

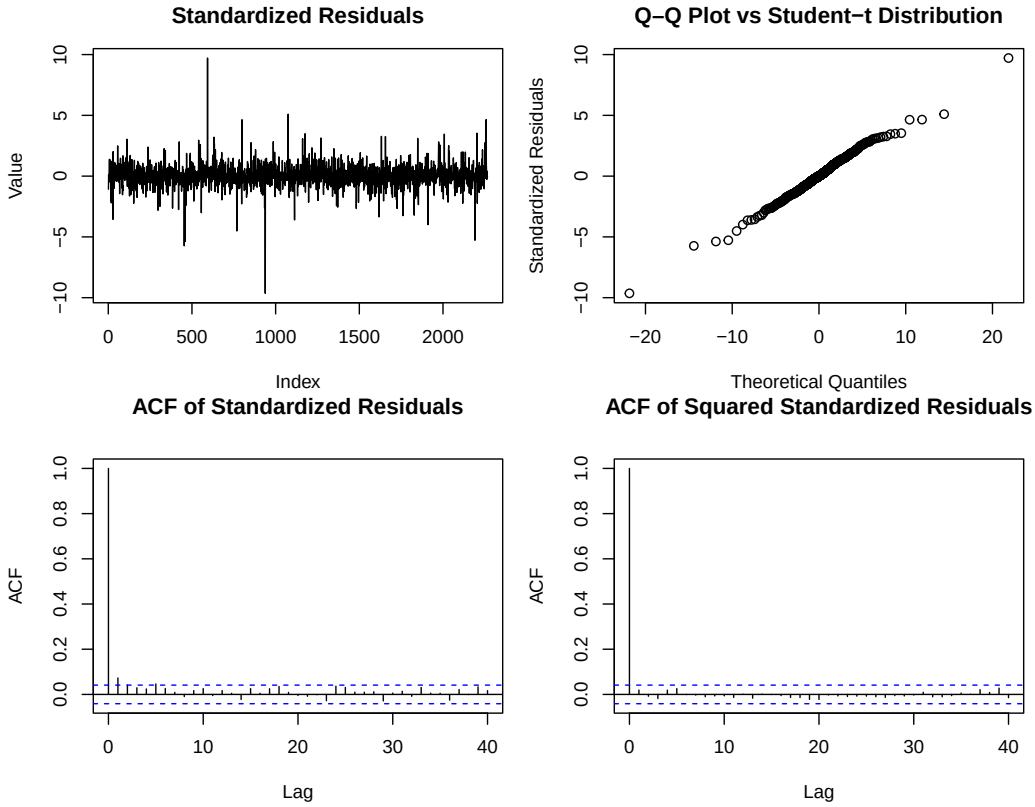


Figure A13: Residual Diagnostics based on z_t for BTC Daily EGARCH ARMA-GARCH(1,0;1,1)

Parameter	Estimate	Std. Error	t value	Pr(>|t|)
μ	0.000225	0.000082	2.737	0.0062
AR(1)	-0.048725	0.008131	-5.993	< 0.0001
ω	0.018735	0.003108	6.028	< 0.0001
α_1	-0.096022	0.065350	-1.469	0.1417
α_2	-0.042871	0.017407	-2.463	0.0138
α_3	0.053673	0.062099	0.864	0.3874
β_1	-0.239330	0.000011	-21046.0	< 0.0001
β_2	1.000000	0.000009	112150.0	< 0.0001
β_3	0.239330	0.000011	20815.0	< 0.0001
γ_1	0.641297	0.000288	2225.3	< 0.0001
γ_2	0.450065	0.006049	74.407	< 0.0001
γ_3	-0.213667	0.001797	-118.89	< 0.0001
Shape	2.100000	0.002969	707.40	< 0.0001
Model Fit				
Log-Likelihood (ℓ)	25681.54			
Number of Parameters (k)	13			
AIC	-51350.08			
BIC	-51244.64			
Diagnostic p-values				
McLeod-Li (squared residuals)	0.177			
Ljung-Box (standard residuals)	0.000			

Table A14: BTC 6-Hour EGARCH ARMA-GARCH(1,0;3,3) In-Sample Estimation and Diagnostics (Robust Standard Errors)

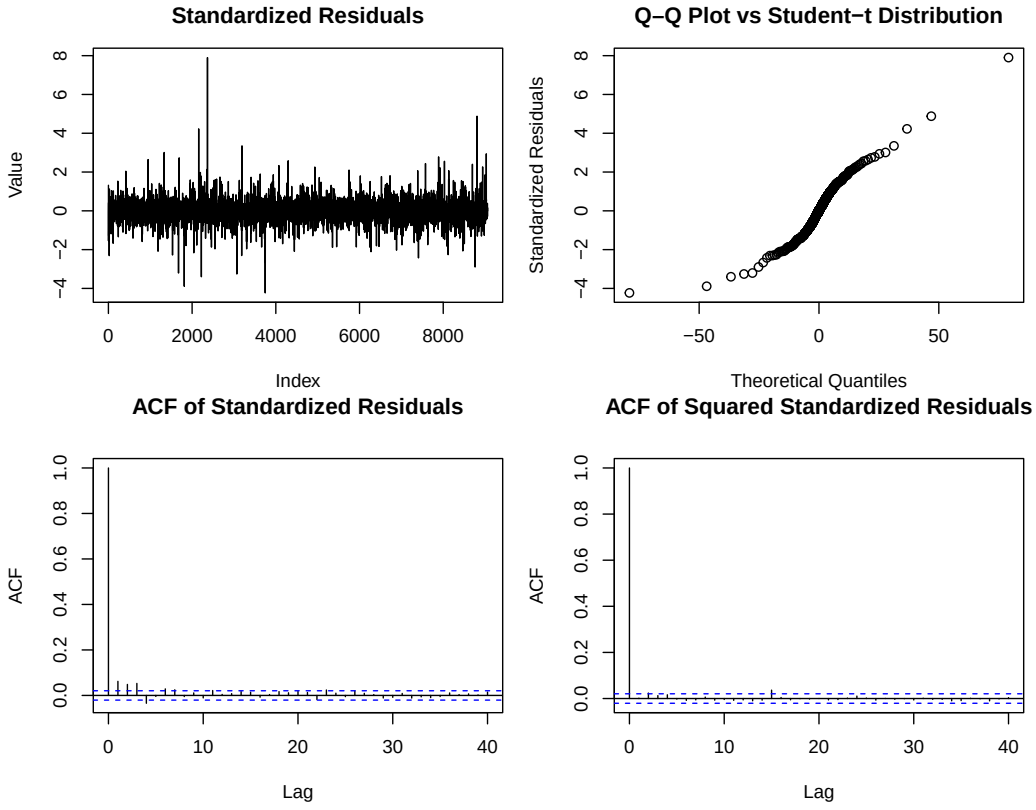


Figure A14: Residual Diagnostics based on z_t for BTC 6-Hour EGARCH ARMA-GARCH(1,0;3,3)

Parameter	Estimate	Std. Error	t value	Pr(>|t|)
μ	0.000754	0.000701	1.076	0.2820
AR(1)	-0.096577	0.018798	-5.138	< 0.0001
ω	-0.077420	0.040093	-1.931	0.0535
α_1	0.002876	0.016910	0.170	0.8649
β_1	0.986870	0.007212	136.838	< 0.0001
γ_1	0.194302	0.076089	2.554	0.0107
Shape	3.174225	0.276204	11.492	< 0.0001
Model Fit				
Log-Likelihood (ℓ)	3966.136			
Number of Parameters (k)	7			
AIC	-7925.272			
BIC	-7878.192			
Diagnostic p-values				
McLeod-Li (squared residuals)	0.557			
Ljung-Box (standard residuals)	0.0016			

Table A15: ETH Daily EGARCH ARMA-GARCH(1,0;1,1) In-Sample Estimation and Diagnostics (Robust Standard Errors)

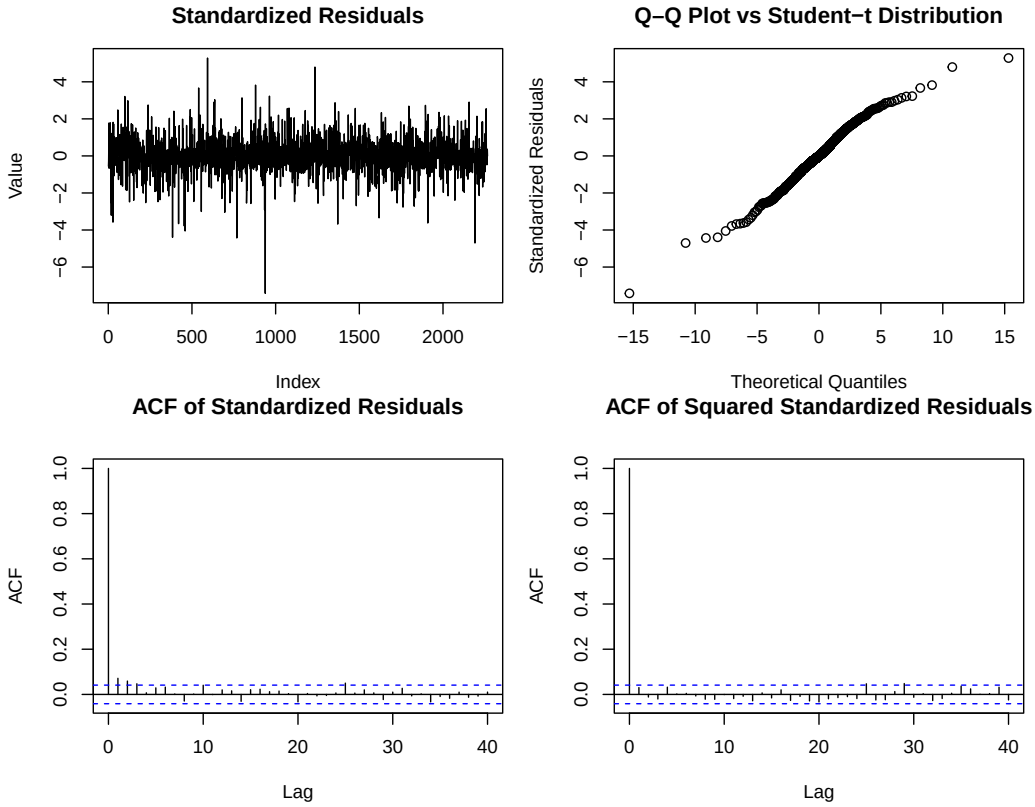


Figure A15: Residual Diagnostics based on z_t for ETH Daily EGARCH ARMA-GARCH(1,0;1,1)

Parameter	Estimate	Std. Error	t value	Pr(>|t|)
μ	0.000117	0.000077	1.528	0.1265
AR(1)	-0.040302	0.009182	-4.389	< 0.0001
ω	-0.117769	0.003322	-35.447	< 0.0001
α_1	-0.021996	0.007018	-3.134	0.0017
α_2	0.001268	0.002984	0.425	0.6708
α_3	-0.022364	0.006533	-3.423	0.0006
β_1	0.991608	0.000000	2.400e6	< 0.0001
β_2	-0.999016	0.000000	-4.636e6	< 0.0001
β_3	0.991064	0.000001	1.823e6	< 0.0001
γ_1	0.228265	0.004380	52.116	< 0.0001
γ_2	-0.004242	0.002997	-1.415	0.1570
γ_3	0.246227	0.005113	48.155	< 0.0001
Shape	2.860369	0.090833	31.490	< 0.0001
Model Fit				
Log-Likelihood (ℓ)	23224.74			
Number of Parameters (k)	13			
AIC	-46436.47			
BIC	-46331.02			
Diagnostic p-values				
McLeod-Li (squared residuals)	0.044			
Ljung-Box (standard residuals)	0.000			

Table A16: ETH 6-Hour EGARCH ARMA-GARCH(1,0;3,3) In-Sample Estimation and Diagnostics (Robust Standard Errors)

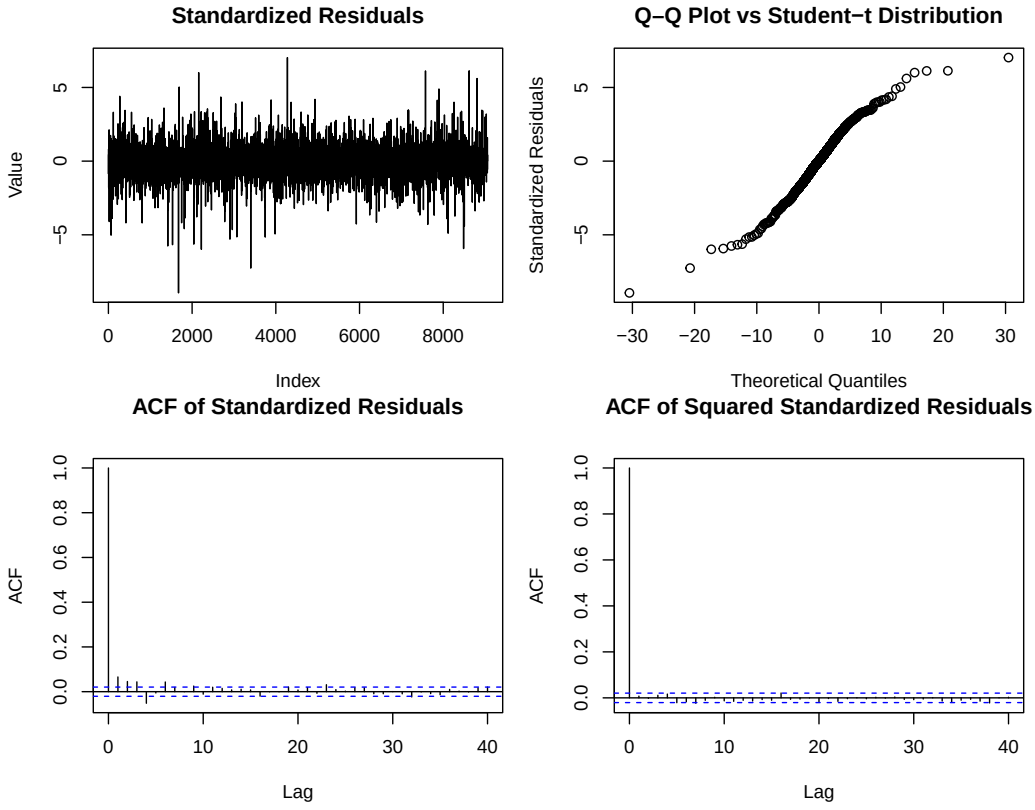


Figure A16: Residual Diagnostics based on z_t for ETH 6-Hour EGARCH ARMA-GARCH(1,0;3,3)

Parameter	Estimate	Std. Error	t value	Pr(>|t|)
$\alpha_0^{(1)}$	-0.0927	0.0165	-5.602	< 0.0001
$\alpha_1^{(1)}$	0.0529	0.0361	1.468	0.0711
$\alpha_2^{(1)}$	0.0360	0.0242	1.490	0.0682
$\beta^{(1)}$	0.9952	0.0012	844.633	< 0.0001
$\nu^{(1)}$	99.5178	0.3752	265.244	< 0.0001
$\alpha_0^{(2)}$	-0.1849	0.2400	-0.771	0.2205
$\alpha_1^{(2)}$	0.2055	0.0824	2.493	0.0063
$\alpha_2^{(2)}$	-0.0238	0.0389	-0.611	0.2706
$\beta^{(2)}$	0.9641	0.0486	19.825	< 0.0001
$\nu^{(2)}$	5.0386	2.7485	1.833	0.0334
P_{11}	0.4425	0.3180	1.392	0.0820
P_{21}	0.3658	0.3552	1.030	0.1516
Model Fit				
Log-Likelihood (ℓ)	4591.159			
Number of Parameters (k)	12			
AIC	-9158.318			
BIC	-9089.608			
Diagnostic p-values				
McLeod-Li (squared residuals)	0.9999			
Ljung-Box (standard residuals)	0.2290			

Table A17: BTC Daily MS-GARCH(1,1) In-Sample Estimation and Diagnostics (Robust Standard Errors)

Regime	To Regime 1	To Regime 2
From Regime 1	0.4425	0.5575
From Regime 2	0.3658	0.6342

Table A18: Transition Probability Matrix for BTC Daily MS-GARCH(1,1)

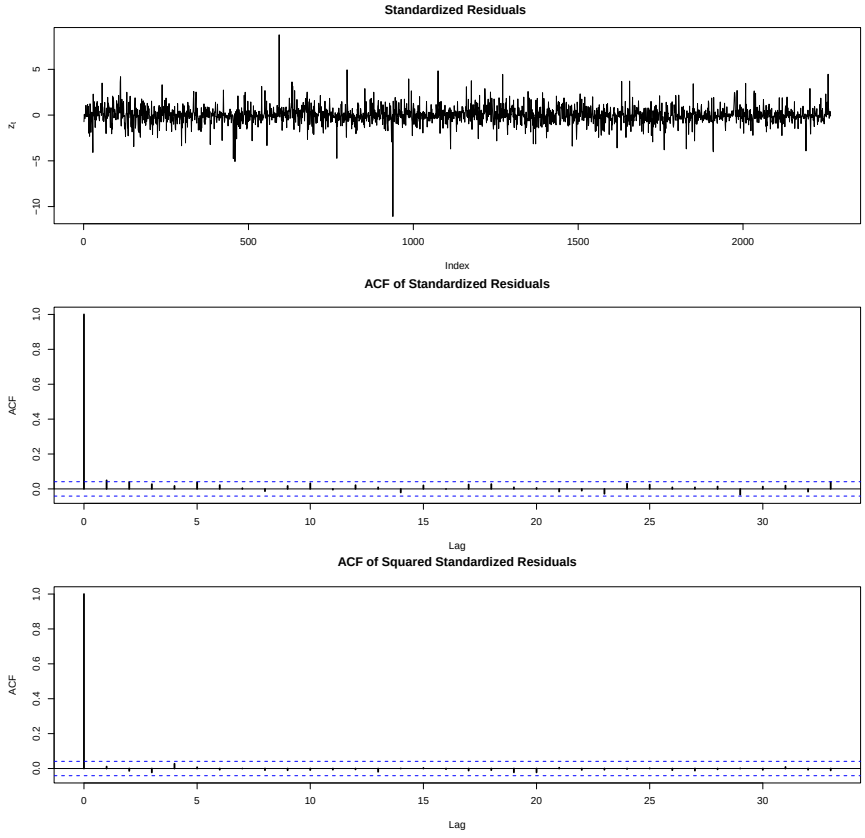


Figure A17: Residual Diagnostics based on z_t for BTC Daily MS-GARCH(1,1)

Parameter	Estimate	Std. Error	t value	Pr(>|t|)
$\alpha_0^{(1)}$	-0.0377	0.0039	-9.751	< 0.0001
$\alpha_1^{(1)}$	0.1008	0.0071	14.142	< 0.0001
$\alpha_2^{(1)}$	-0.0005	0.0071	-0.074	0.4705
$\beta^{(1)}$	0.9977	0.0002	4042.764	< 0.0001
$\nu^{(1)}$	3.0168	0.0310	97.357	< 0.0001
$\alpha_0^{(2)}$	-0.3384	0.0154	-21.939	< 0.0001
$\alpha_1^{(2)}$	0.2541	0.0246	10.321	< 0.0001
$\alpha_2^{(2)}$	-0.0759	0.0207	-3.672	0.0001
$\beta^{(2)}$	0.9419	0.0018	522.066	< 0.0001
$\nu^{(2)}$	6.7383	0.1493	45.140	< 0.0001
P_{11}	0.8713	0.0085	102.981	< 0.0001
P_{21}	0.5067	0.0038	134.523	< 0.0001
Model Fit				
Log-Likelihood (ℓ)	25751.91			
Number of Parameters (k)	12			
AIC	-51479.81			
BIC	-51394.48			
Diagnostic p-values				
McLeod-Li (squared residuals)	0.939			
Ljung-Box (standard residuals)	5.95×10^{-13}			

Table A19: BTC 6-Hour MS-GARCH(1,1) In-Sample Estimation and Diagnostics (Robust Standard Errors)

Regime	To Regime 1	To Regime 2
From Regime 1	0.8713	0.1287
From Regime 2	0.5067	0.4933

Table A20: Transition Probability Matrix for BTC 6-Hour MS-GARCH(1,1)

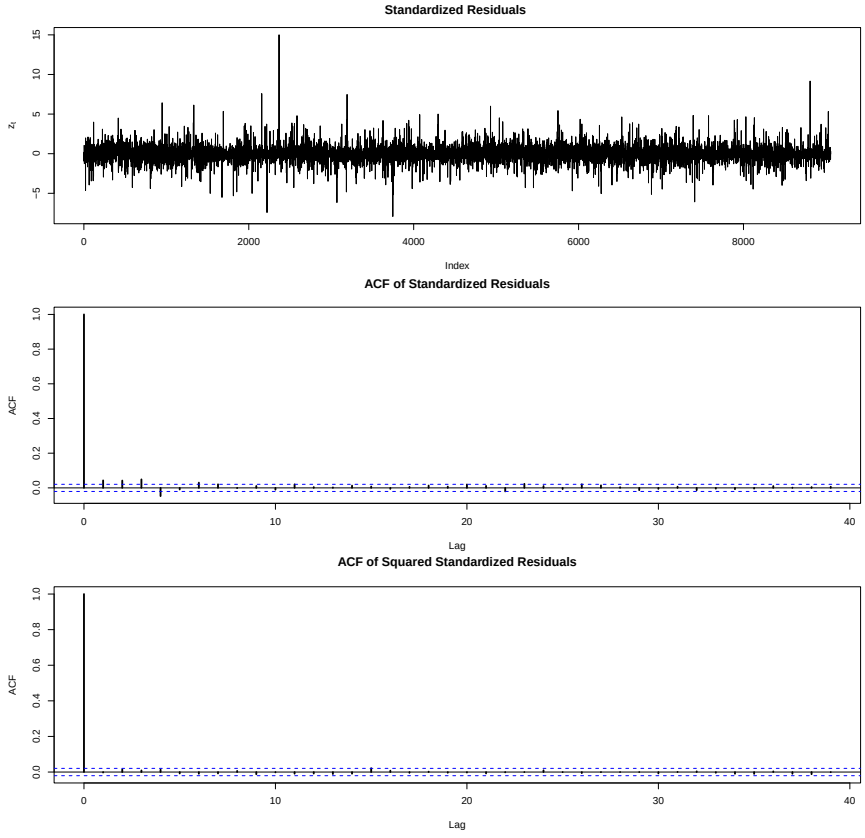


Figure A18: Residual Diagnostics based on z_t for BTC 6-Hour MS-GARCH(1,1)

Parameter	Estimate	Std. Error	t value	Pr(>|t|)
$\alpha_0^{(1)}$	-0.0514	0.0298	-1.727	0.0421
$\alpha_1^{(1)}$	0.2072	0.0419	4.940	< 0.0001
$\alpha_2^{(1)}$	0.0597	0.0309	1.934	0.0266
$\beta^{(1)}$	0.9934	0.0039	251.992	< 0.0001
$\nu^{(1)}$	3.0589	0.4066	7.523	< 0.0001
$\alpha_0^{(2)}$	-1.6559	1.0258	-1.614	0.0532
$\alpha_1^{(2)}$	0.1263	0.1149	1.100	0.1358
$\alpha_2^{(2)}$	-0.3379	0.0781	-4.325	< 0.0001
$\beta^{(2)}$	0.7211	0.1785	4.039	< 0.0001
$\nu^{(2)}$	35.4027	105.0627	0.337	0.3681
P_{11}	0.8304	0.1008	8.239	< 0.0001
P_{21}	0.5430	0.0891	6.095	< 0.0001
Model Fit				
Log-Likelihood (ℓ)	3974.884			
Number of Parameters (k)	12			
AIC	-7925.767			
BIC	-7857.058			
Diagnostic p-values				
McLeod-Li (squared residuals)	0.210			
Ljung-Box (standard residuals)	0.061			

Table A21: ETH Daily MS-GARCH(1,1) In-Sample Estimation and Diagnostics (Robust Standard Errors)

Regime	To Regime 1	To Regime 2
From Regime 1	0.8304	0.1696
From Regime 2	0.5430	0.4570

Table A22: Transition Probability Matrix for ETH Daily MS-GARCH(1,1)

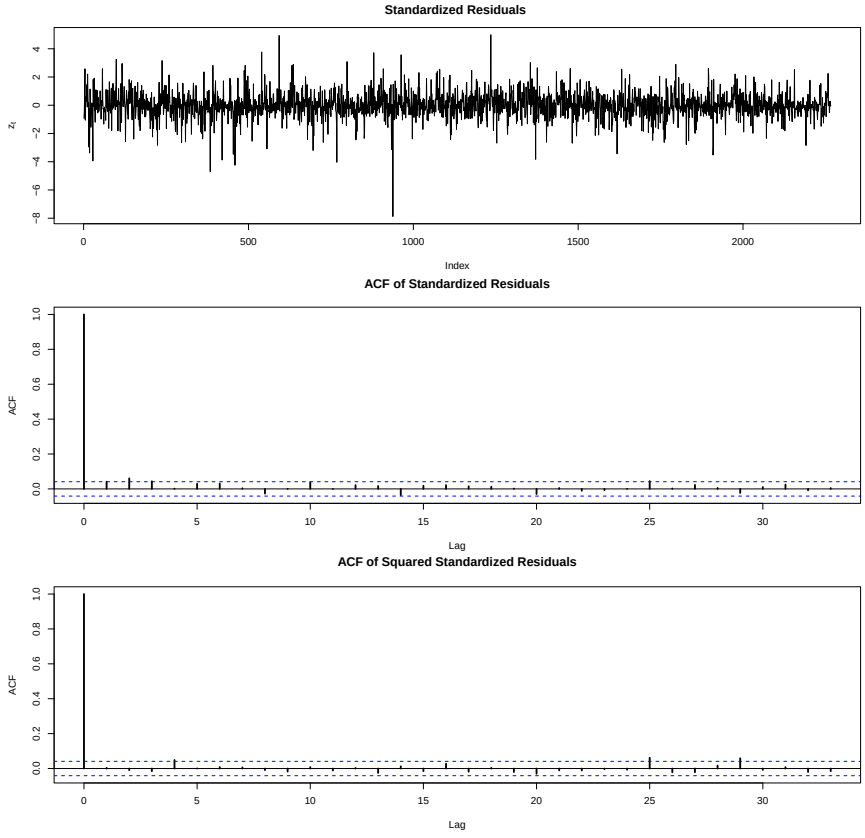


Figure A19: Residual Diagnostics based on z_t for ETH Daily MS-GARCH(1,1)

Parameter	Estimate	Std. Error	t value	Pr(>|t|)
$\alpha_0^{(1)}$	-0.0342	0.0036	-9.510	< 0.0001
$\alpha_1^{(1)}$	0.1048	0.0080	13.101	< 0.0001
$\alpha_2^{(1)}$	0.0019	0.0095	0.197	0.4220
$\beta^{(1)}$	0.9980	0.0002	4410.468	< 0.0001
$\nu^{(1)}$	3.3834	0.0444	76.154	< 0.0001
$\alpha_0^{(2)}$	-0.2156	0.0103	-20.957	< 0.0001
$\alpha_1^{(2)}$	0.1647	0.0192	8.587	< 0.0001
$\alpha_2^{(2)}$	-0.0540	0.0150	-3.596	< 0.0001
$\beta^{(2)}$	0.9624	0.0013	766.162	< 0.0001
$\nu^{(2)}$	7.2561	0.1714	42.331	< 0.0001
P_{11}	0.9178	0.0066	138.683	< 0.0001
P_{21}	0.2463	0.0026	93.576	< 0.0001
Model Fit				
Log-Likelihood (ℓ)	23236.73			
Number of Parameters (k)	12			
AIC	-46449.46			
BIC	-46364.12			
Diagnostic p-values				
McLeod-Li (squared residuals)	0.163			
Ljung-Box (standard residuals)	3.33×10^{-16}			

Table A23: ETH 6-Hour MS-GARCH(1,1) In-Sample Estimation and Diagnostics (Robust Standard Errors)

Regime	To Regime 1	To Regime 2
From Regime 1	0.9178	0.0822
From Regime 2	0.2463	0.7537

Table A24: Transition Probability Matrix for ETH 6-Hour MS-GARCH(1,1)

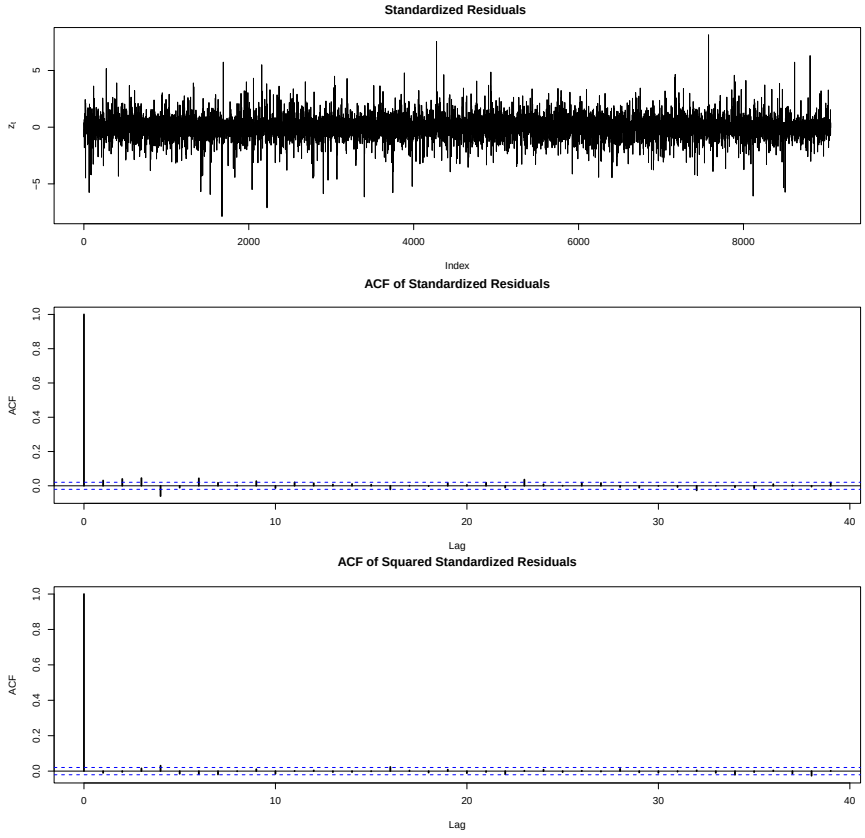


Figure A20: Residual Diagnostics based on z_t for ETH 6-Hour MS-GARCH(1,1)

Parameter	Estimate	Std. Error	t value	Pr($>\|t\|$)
BTC Series				
[BTC] μ	0.000204	0.000076	2.701	0.0069
[BTC] AR(1)	-0.050055	0.010293	-4.863	< 0.0001
[BTC] ω	0.004368	0.000691	6.321	< 0.0001
[BTC] α_1	-0.100282	0.036689	-2.733	0.0063
[BTC] α_2	0.004266	0.001799	2.371	0.0177
[BTC] α_3	0.089770	0.034760	2.583	0.0098
[BTC] β_1	0.675589	0.000000	5616900.0	< 0.0001
[BTC] β_2	0.994473	0.000090	11050.0	< 0.0001
[BTC] β_3	-0.670063	0.000000	-84164000.0	< 0.0001
[BTC] γ_1	0.661156	0.055260	11.965	< 0.0001
[BTC] γ_2	0.131982	0.009346	14.122	< 0.0001
[BTC] γ_3	-0.592827	0.053377	-11.107	< 0.0001
[BTC] ν	2.100000	0.004409	476.260	< 0.0001
ETH Series				
[ETH] μ	0.000139	0.000053	2.644	0.0082
[ETH] AR(1)	-0.037522	0.007705	-4.870	< 0.0001
[ETH] ω	-0.056135	0.002591	-21.664	< 0.0001
[ETH] α_1	-0.024502	0.001358	-18.043	< 0.0001
[ETH] β_1	0.992141	0.000047	21085.0	< 0.0001
[ETH] γ_1	0.235281	0.019474	12.082	< 0.0001
[ETH] ν	2.812688	0.102934	27.325	< 0.0001
DCC Parameters				
$dcca_1$	0.030630	0.003706	8.265	< 0.0001
$dccb_1$	0.943982	0.006843	137.950	< 0.0001
mshape (Joint ν)	4.000000	0.043195	92.604	< 0.0001
Model Fit				
Log-Likelihood (ℓ)		48594.32		
Number of Parameters (k)		24		
AIC		-97165.65		
BIC		-97010.96		

Table A25: 6-Hour AR(1)-eGARCH DCC-GARCH(3,3) Estimation with Student-t Innovations

Parameter	Estimate	Std. Error	t value	Pr($>\|t\|$)
BTC Series				
[BTC] μ	0.000043	0.000011	4.047	0.0001
[BTC] AR(1)	-0.091264	0.003842	-23.753	< 0.0001
[BTC] ω	-0.007171	0.000187	-38.418	< 0.0001
[BTC] α_1	-0.037157	0.001254	-29.631	< 0.0001
[BTC] α_2	-0.003357	0.000958	-3.505	0.0005
[BTC] α_3	0.036293	0.002143	16.935	< 0.0001
[BTC] β_1	0.628703	0.000000	4902300.0	< 0.0001
[BTC] β_2	0.998455	0.000011	89900.0	< 0.0001
[BTC] β_3	-0.627872	0.000009	-69394000.0	< 0.0001
[BTC] γ_1	0.359378	0.000039	9232.9	< 0.0001
[BTC] γ_2	0.044018	0.000995	44.231	< 0.0001
[BTC] γ_3	-0.317247	0.000027	-11696.0	< 0.0001
[BTC] ν	2.792370	0.003593	777.210	< 0.0001
ETH Series				
[ETH] μ	0.000022	0.000017	1.304	0.1923
[ETH] AR(1)	-0.075753	0.004216	-17.967	< 0.0001
[ETH] ω	-0.040590	0.000500	-81.160	< 0.0001
[ETH] α_1	-0.022050	0.002694	-8.185	< 0.0001
[ETH] β_1	0.995704	0.000033	30541.0	< 0.0001
[ETH] γ_1	0.172972	0.002082	83.076	< 0.0001
[ETH] ν	2.930480	0.063015	46.505	< 0.0001
DCC Parameters				
$dcca_1$	0.026648	0.001345	19.819	< 0.0001
$dccb_1$	0.972383	0.001443	673.640	< 0.0001
mshape (Joint ν)	4.000000	0.074832	53.453	< 0.0001
Model Fit				
Log-Likelihood (ℓ)		432632.5		
Number of Parameters (k)		24		
AIC		-865241.9		
BIC		-865087.2		

Table A26: Hourly AR(1)-eGARCH DCC-GARCH(3,3) Estimation with Student-t Innovations